

Abstract

LDOs gets popular in integrated circuit design including functions such as reducing switching ripples from high efficiency regulators, cancelling spurs from other loads, and giving different supply voltages to loads. In accordance with load applications, choosing proper LDOs is important. Analog loads such as voltage-controlled oscillators and analog-to-digital converters need LDOs that have high power-supply-rejection-ratio (PSRR), high accuracy, and low noise.

The major performance highlight is the high PSRR measured from the Cadence response: approximately -80.95 dB at 1 kHz, -80.76 dB at 10 kHz, -80.29 dB at 100 kHz, and around -64.26 dB at 1 MHz in the plotted result. The loop gain and phase plot shows a phase margin of about 66.86° and gain margin of about 79.02 dB, while transient results indicate approximately 0.5 mV overshoot/undershoot and a 5 μ s settling time. These results show that the proposed LDO is suitable for low-noise wearable health-monitoring applications

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CHAPTER 1

1.1 INTRODUCTION:

A Low Dropout Regulator (LDO) is a fundamental analog building block widely used in modern electronic systems to provide a stable and noise-free output voltage despite variations in input supply and load conditions. In wearable devices—such as fitness trackers, smartwatches, and biomedical sensors—power management becomes critically important due to strict constraints on battery life, compact size, and continuous operation. These systems often integrate sensitive analog and mixed-signal circuits like ADCs, RF modules, and biosignal amplifiers, all of which demand a clean and stable supply. The LDO plays a vital role in ensuring voltage regulation with minimal ripple, low dropout voltage, and high efficiency, thereby directly impacting the reliability and longevity of wearable electronics.

Wearable health monitoring devices such as ECG patches, smart watches, pulse monitoring nodes, glucose sensors and portable diagnostic modules depend on a reliable power management unit. The analog front-end in these systems must process microvolt- and millivolt-level biological signals, so supply ripple can directly appear as measurement error, baseline drift or false detection. LDO regulators are therefore used after batteries or switching converters to provide a quiet local supply for sensitive analog, mixed-signal and RF blocks [4], [5].

An LDO is preferred in these applications because it is simpler and quieter than a switching converter, operates with a small input-output voltage headroom and offers good output accuracy when the feedback loop is properly designed. The price paid for this simplicity is that PSRR, dropout voltage, transient response and stability become strongly dependent on the pass transistor, error amplifier gain, compensation network and load conditions [6], [11].

The present project focuses on a high-PSRR LDO implemented in Cadence Virtuoso. The goal is not only to obtain a regulated output voltage, but to demonstrate measurable improvement in line regulation, load regulation, PSRR, loop stability and transient behavior. This makes the work suitable for wearable health monitoring devices where power integrity and long battery life must be considered together [5], [7].

For the mobile and internet-of-things (IoT) applications, the compact integration and low power dissipation is necessary in chip design. System-on-chip (SoC) that is contained every function in a compact chip become popular since it can remove chip-to-chip path-caused parasitic by removing power and signal delivery between chips. According this tendency, power management integrated circuits (PMICs) should be fully-integrated and thus more complex strategies are needed when designing these circuits.

As shown in the top of Figure 1.1, a battery provides charges to the inputs of DC/DC converters and these converters deliver supply voltages to loads containing charges in capacitors ($C_{L,DC/DC}$) as the level of desired voltage. However, these supplies contain large switching ripples by the operation properties of these converters and spurs from other loads sharing same supply voltage. These problems make the loads which need clean and noise-less supply cannot operate properly. Furthermore, the loads can operate optimally in their own supply voltage level but DC/DC converters only provide same voltage level to their loads.

To resolve these problems, low-dropout voltage regulators (LDOs) are added between switching converters and loads as shown in the bottom of Figure 1. Even if the LDOs have lower power efficiency than DC/DC converters due to series connection on current provision path, they can suppress switching ripples from DC/DC converters, reduce spurs from other loads, and provide different supply voltage to the different loads.

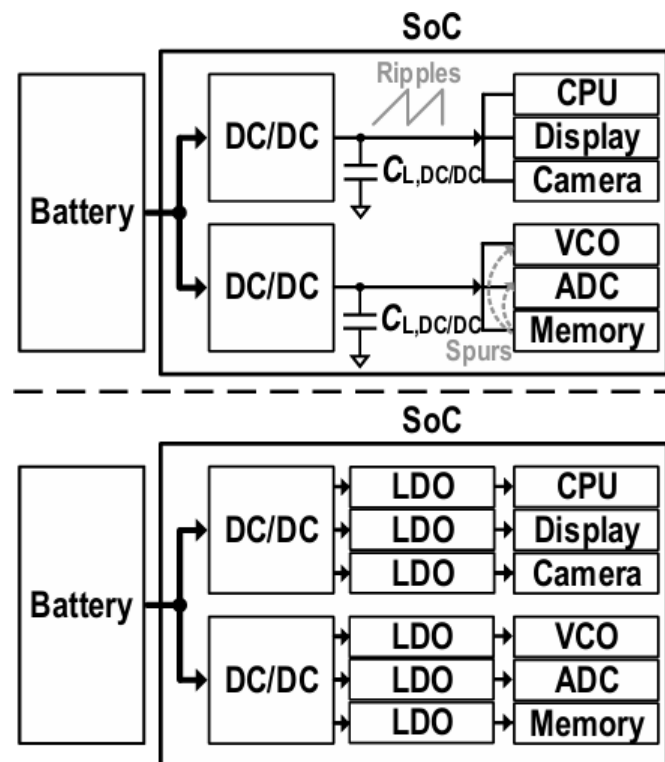


Figure 1.1 The path of charge delivery from a battery to loads without LDOs (top) and with LDOs

1.1.1 Operation of LDOs

As shown in Figure 1.2, analog LDOs basically consist of a P-type pass transistor (MP), an error amplifier (EA), and a load capacitor (CL). If I_L or V_{IN} changes, CL compensates the change firstly occurring fluctuation in V_{OUT} . In the second step, feedback loop operates to regulate MP current (I_{OUT}) as the same value of I_L , restoring V_{OUT} same value with reference voltage (V_{REF}).

For example, as shown in the top of Figure 1.3, when I_L changes from minimum I_L ($I_{L,min}$) to maximum I_L ($I_{L,max}$), the current from CL (I_{CL}) initially compensates I_L , and it evokes V_{OUT} decrease. After that, the negative feedback loop operates: input difference of EA decreases by V_{OUT} decreasing and the reduction is amplified with EA gain (AEA), evoking huge decrease of the gate node voltage (V_G) of MP. This increase of $|V_{GS}|$ of MP results in huge increase of I_{OUT} , so MP can provide current in the same quantity with I_L , and thus put V_{OUT} on almost same value with V_{REF} . As shown in the bottom of Figure 2, when I_{CL} compensates I_L initially, V_{OUT} decreases, but as the portion of I_{OUT} –compensation increases, V_{OUT} gets back to the value of V_{REF} .

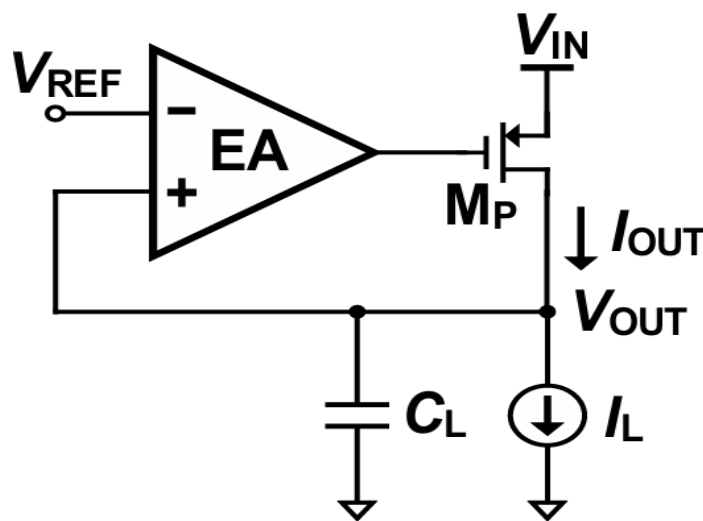
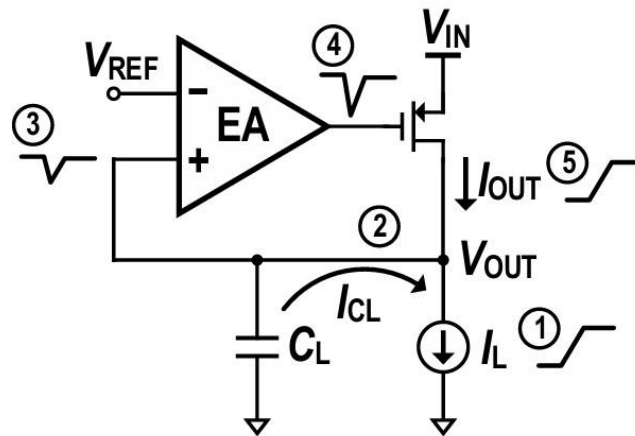


Figure 1.2 Basic structure of an LDO



V_{OUT} @load transient

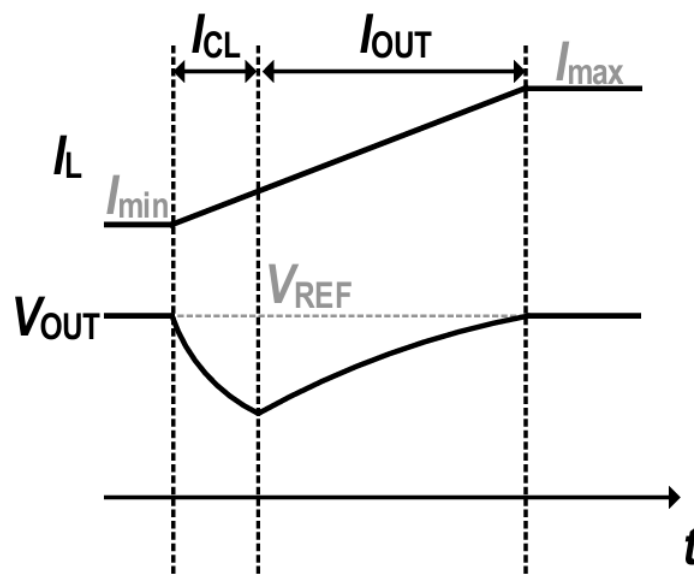


Figure 1.3 LDO operation when I_L changes from $I_{L,min}$ to $I_{L,max}$.

1.1.2 Types of Pass Transistor

We need to select the type of pass transistor between P-type and N-type for desired purpose. M_P has smaller dropout voltage (V_{DO}), lower power supply rejection ratio (PSRR), and lower output pole frequency ($\omega_{P,OUT}$) than N-type pass transistor (M_N). The comparison of pass transistor type is shown in Table 1.

V_{DO} is the dropout voltage between V_{IN} and V_{OUT} . As shown in Figure 1.4, For M_N operating in saturation region, V_{DO} should be large. Otherwise, V_G should be larger than V_{IN} . Because V_G should be larger than the sum of V_{OUT} and threshold voltage (V_{th}) of M_N . For example, if V_{th} of M_N is over 300mV and the specification needs under 200mV V_{DO} , charge pump is needed for increasing V_G as larger than $V_{IN} + 100\text{mV}$.

PSRR is the ratio of the change in V_{IN} to the change in V_{OUT} it produces. If V_{IN} fluctuates, in M_N case, V_{OUT} is robust to V_{IN} fluctuation since in view of V_{OUT} , only drain of M_N fluctuates and MOSFET is inherently robust to the change of V_{DS} . However, M_P amplifying as M_P gain (A_{MP}) and deliver to V_{OUT} since the change occurs in V_{GS} . Thus, we generally say M_N type LDOs have high PSRR than M_P type LDOs. However, it is just a part of tendency and if deep analysis is conducted, the solution will be ambiguous. In brief, after determining the type of pass transistor, EA type should be selected as the way v_R cannot be represented in V_{OUT} .

General structure of LDO has two low frequency poles. $\omega_{P,OUT}$ is the pole at the node of V_{OUT} , and $\omega_{P,G}$ is the pole at the node of V_G . In M_N case, since low $1/g_m$ impedance of M_N is seen by V_{OUT} , $\omega_{P,OUT}$ can easily be higher than $\omega_{P,G}$ and makes LDO gate-pole-dominant (GPD). However, in M_P case, since the high r_O impedance of M_P is seen by V_{OUT} , it can be negligible, so by the condition of load impedance, it can be GPD or output-pole-dominant (OPD) LDO. The characteristics of GPD and OPD LDO are explained in Section 1.1.3.

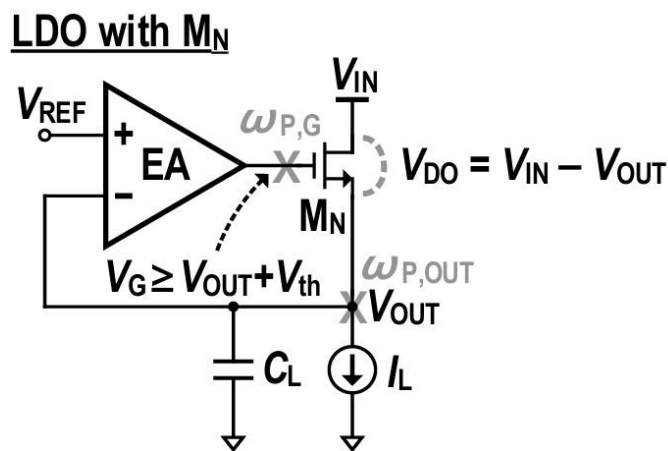


Figure 1.4 Analog LDO with M

Table 1.1 Comparison between MP and MN

Pass transistor type	P-type (M_P)	N-type (M_N)
V_{DO}	Low	High
PSRR	Low	High
$\omega_{P,OUT}$	Depend on I_L	High

1.1.3 Location of Dominant Pole

Location of dominant pole is important consideration. In the past years, almost all LDOs were OPD since designers use large off-chip capacitor so $\omega_{P,OUT}$ is near to DC making feedback loop stable even in the situation that $\omega_{P,G}$ is somewhat small. Moreover, this large C_L can bypass the noise of LDOs, and can be robust to load transition since the large C_L compensates most I_L with little fluctuation of V_{OUT} (ΔV_{OUT}).

However, in recent years, fully-integrated LDOs are required to reduce PCB area and the number of pins in chips. As C_L size is reduced, satisfying stability condition becomes difficult in OPD LDOs. In some applications, over 10mA I_L is needed. For these reasons, GPD LDOs get popular for fullyintegrated design. The differences between OPD and GPD LDOs are summarized in Table 2.

Table 1.2 Comparison of GPD and OPD LDOs

Location of dominant pole	Gate	Output
Area	Small	Large
Noise	High	Low
ΔV_{OUT}	High	Low
Stability @ $I_{L,min}$	Poor	Good
Stability @ $I_{L,max}$	Good	Poor

1.2 PERFORMANCE METRICS OF LDOS

1.2.1 Power-Supply Rejection Ratio (PSRR):

DC/DC converter ripple suppression is one of the major objectives of an LDO, especially when the LDO is placed after a switching converter in a power management system. The input ripple can reach the output through different coupling paths in the pass transistor. One path is directly through the finite drain-source resistance of the pass transistor, while another path is through the gate node, where parasitic capacitances such as C_{gs} and C_{gd} influence how much ripple appears at the output. Therefore, reducing the difference between gate and source ripple is important for improving PSRR and maintaining a clean output voltage [7], [8], [10]. The ripple-related output current variation can be expressed as:

$$\Delta I_{OUT} = g_{ds} \cdot v_R + g_m \cdot (v_R - v_R \cdot \frac{C_{gs}}{C_{gs} + C_{gd}}) \quad (1)$$

where g_{ds} is the drain-source conductance of the pass transistor, g_m is the transconductance of the pass transistor, C_{gs} is the gate-source capacitance, and C_{gd} is the gate-drain capacitance. This expression shows that the output ripple depends on both the direct output resistance path and the gate-coupled path of the pass transistor. Hence, high loop gain, proper compensation, and ripple cancellation techniques are important for improving PSRR in LDO design [7], [8].

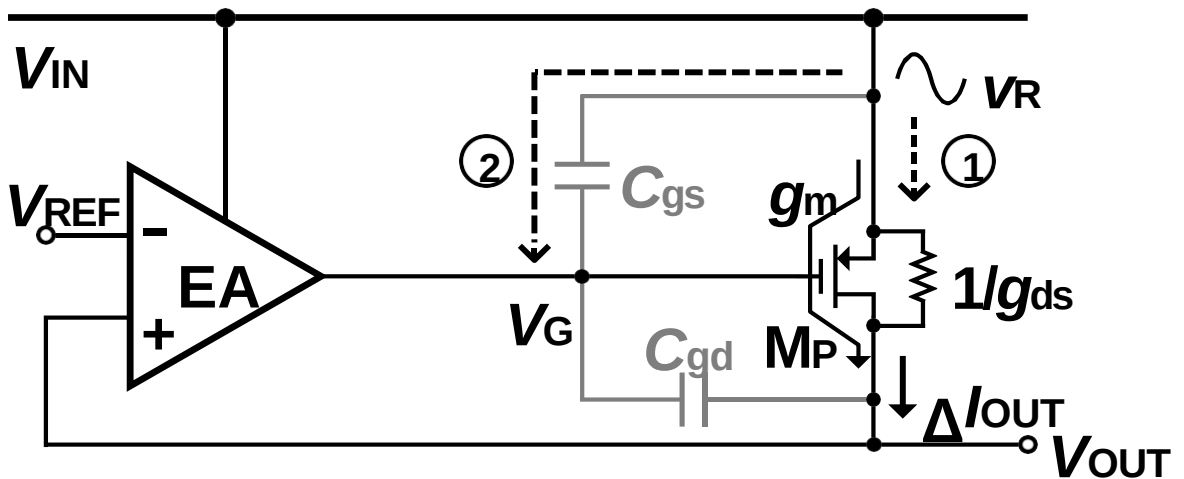


Figure 1.5 The paths that v_R are represented to V_{OUT} .

As mentioned in Section 1.1.2, in the path related to the first term of (1), v_R are multiplied by g_m in the M_P case,

determines the fluctuation of ΔI_{OUT} . In the M_P case, increasing C_{gs} (between V_G and V_{IN}) helps improve PSRR performance and in the M_N case, increasing C_{gs} (between V_G and V_{OUT}) helps improve PSRR performance. Plus, we should consider the situation that v_R affect the output of EA and generates V_G ripples.

As shown in Figure 1.6–1.7, (2), and (3), we can remove these ripples by selecting suitable structure of EA. Briefly, for maximizing v_R to V_G fluctuating, PMOS mirror can be selected as it delivers the current that affects V_G in the same direction of resistive dividing path. In the opposite case, NMOS mirror can affect V_G in the opposite direction of resistive dividing path. It is worth noting that before-mentioned techniques are only applied to improve PSRR in the low frequency band [7], [8].

Figure 1.8 shows PSRR along frequency in the LDO that includes M_P and NMOS input PMOS mirror EA combination. In DC region, PSRR has the value $A_{EA} \cdot A_{MP}$ (A_{OL}). If the frequency over -3dB bandwidth ($BW_{-3\text{dB}}$) of loop that is $\omega_{p,G}$, A_{EA} is reduced and PSRR performance gets degradation. In OPD LDO case, the PSR is kept constant since the load impedance rolls-off not only A_{EA} does. To return to the GPD LDO case, if the frequency over the unity gain frequency (UGF), there is no more gain from EA, so the PSRR is kept constant. If the frequency over the output pole, load impedance rolls off, so the PSRR increases. If the frequency over the $BW_{-3\text{dB}}$ of load impedance that is made by C_L and equivalent series resistance (R_{ESR}) of the C_L , load impedance remains constant, so the PSRR is kept constant [7], [10].

The $BW_{-3\text{dB}}$ of PSRR (p_1) can be more important than $PSRR_{DC}$ in the situation high frequency spur is injected into V_{IN} such as the optical receiver applications. Then, we can flip the combination from $PSRR_{DC}$ desiring combination [4], [5]. The characteristics of combinations are summarized in Table 1.3

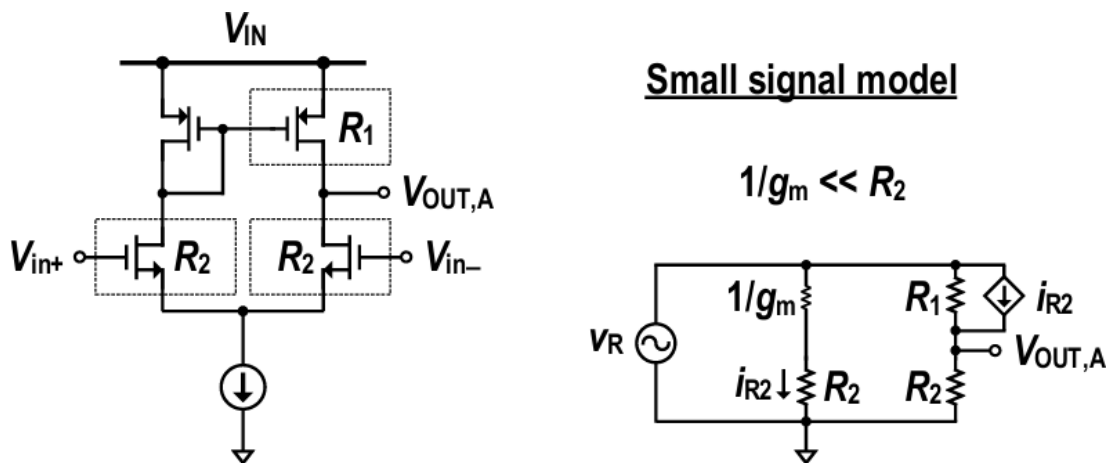
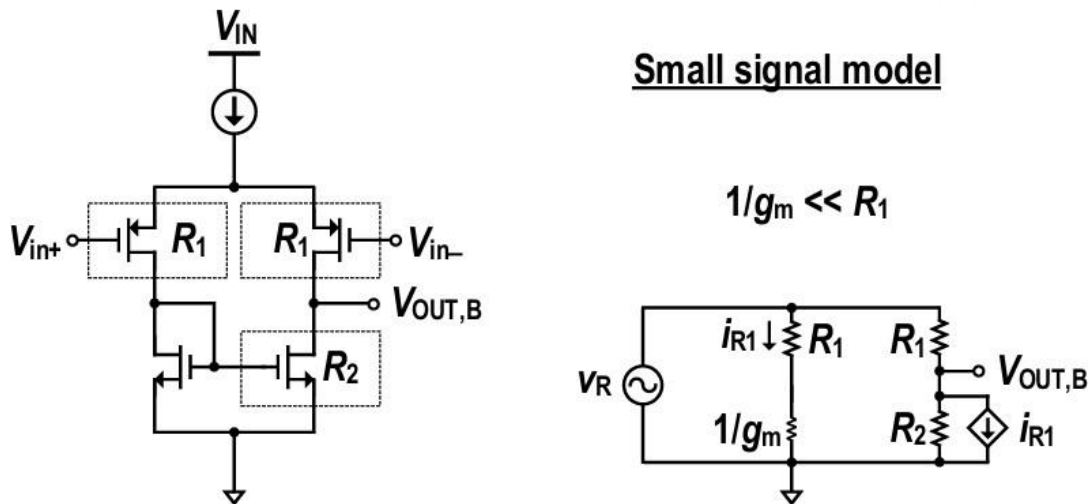


Figure 1.6 Delivering v_R to EA output fully in NMOS-input, PMOS-mirror.

$$V_{out,A} = v_R \cdot R_1 + R_2 R_2 + i_{R2} (R_1 \parallel R_2) \quad (2)$$

$$\approx v_R \cdot \frac{R_2}{R_1 + R_2} + \frac{v_R}{R_2} \left(-\frac{R_1 R_2}{R_1 + R_2} \right) = v_R$$

Figure 1.7 Delivering v_R to EA output scarcely in PMOS-input, NMOS-mirror.

$$V_{OUT,B} = v_R \cdot \frac{R_2}{R_1 + R_2} - i_{R1}(R_1 || R_2)$$

$$\approx v_R \cdot \frac{R_2}{R_1 + R_2} - \frac{v_R}{R_1} \left(\frac{R_1 R_2}{R_1 + R_2} \right) = 0 \quad (3)$$

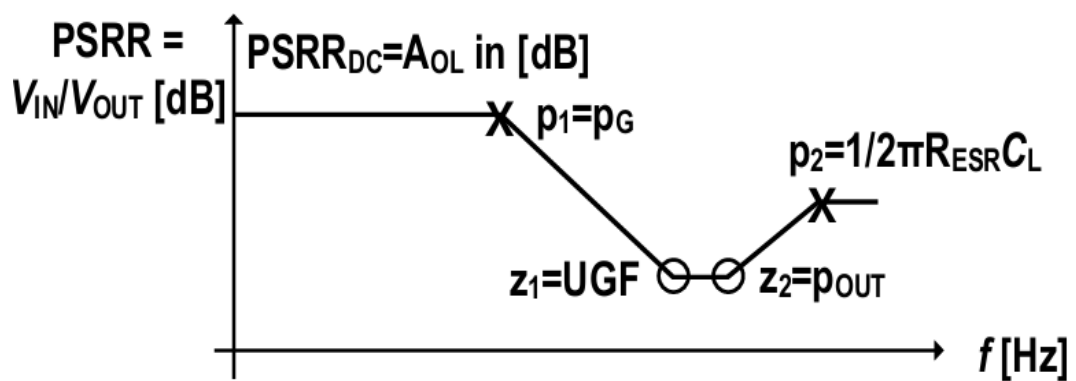


Figure 1.8 PSR change along frequency.

Table 1.3 The characteristics of combinations

Pass transistor type	EA	PSRR _{DC}	p ₁
M _P	NMOS input PMOS mirror	A _{OL}	p _G
	PMOS input NMOS mirror	A _{EA}	A _{MP} p _G
M _N	NMOS input PMOS mirror	A _{EA}	A _{MP} p _G
	PMOS input NMOS mirror	A _{OL}	p _G

1.2.2 Output Voltage Noise ($V_{n,rms}$):

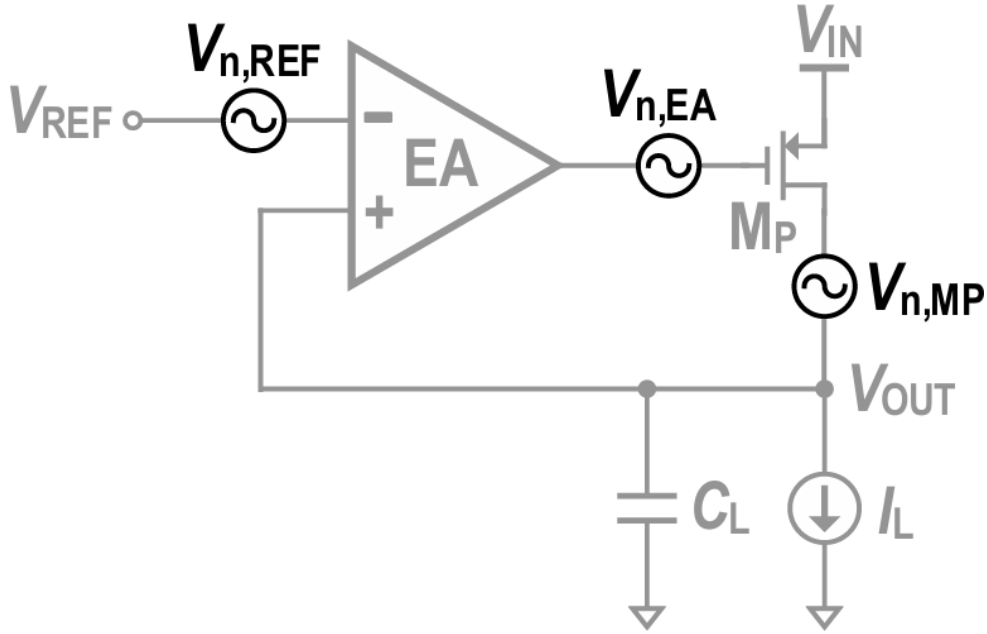


Figure 1.9 Different noise sources of LDOs

Output voltage noise ($V_{n,rms}$) is the total noise at the output node generated by a LDO itself. In some applications such as passive sensors, $V_{n,rms}$ becomes an important factor. Different noise sources are represented as shown in Figure 1.9, V_{REF} noise ($V_{n,REF}$) comes from band-gap-reference (BGR) circuit, EA noise ($V_{n,EA}$) comes from EA, and M_P noise ($V_{n,MP}$) comes from M_P . $V_{n,REF}$ and $V_{n,EA}$ are dominant noise sources since they are amplified passing through the loop. To suppress $V_{n,REF}$, a bypass capacitor at the node of V_{REF} can be added. An effective way to reduce $V_{n,rms}$ is to increase the value of C_L sacrificing the chip area. The types of noise sources are thermal noise and flicker noises. We can calculate the value with

$$\begin{aligned} \overline{V_{n,thermal}^2} &= 4kTR \cdot \frac{1}{1 + \frac{s}{\omega_p}} \\ \overline{V_{n,flicker}^2} &= \frac{K}{C_{ox}WL} \cdot \frac{1}{f} \end{aligned} \quad (4)$$

where k is Boltzmann's constant, T is the resistors' absolute temperature, R is resistance, K is the process-dependent constant, C_{ox} is the oxide capacitance in MOSFET devices, and W and L is channel width and length, respectively.

1.2.3 Line & Load Regulation:

The line and load regulation are important specifications for LDO providing exact the level of V_{OUT} . Although the V_{IN} or I_L changes, V_{OUT} should be kept as the same value with V_{REF} . In the cover range of LDOs, the steady-state difference between V_{OUT} and V_{REF} by changing V_{IN} and I_L can be measured and related performance metrics are called line and load regulation, respectively. If a system needs dynamic voltage and frequency scaling (DVFS), line and load regulation of the LDO should be fine to provide desired levels. Some DVFS chips need <10mV resolution. The line and load regulation performance of LDO using M_P can be increased by increasing A_{OL} as shown in (5) and (6).

$$\begin{aligned} \text{Line regulation} &= \frac{\Delta V_{OUT}}{\Delta V_{IN}} = \frac{\left(\frac{r_{ds} || R_L}{A_{EA} A_{MP}} \right)}{r_{ds} + \left(\frac{r_{ds} || R_L}{A_{EA} A_{MP}} \right)} \\ &\approx \frac{\frac{R_L}{A_{OL}}}{r_{ds} + \frac{R_L}{A_{OL}}} \approx \frac{R_L}{r_{ds} A_{OL}} \end{aligned} \quad (5)$$

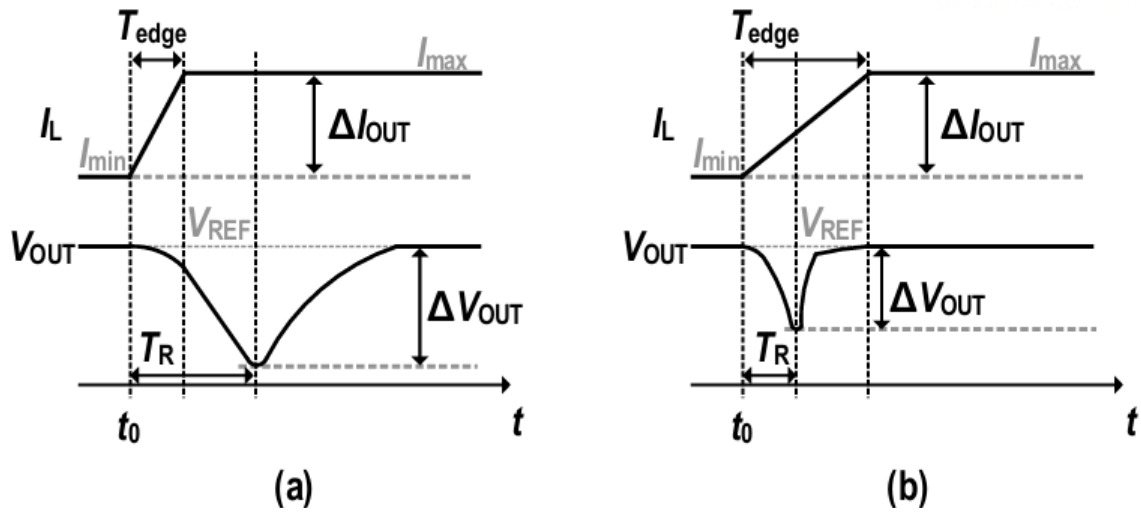
$$\text{Load regulation} = \frac{\Delta V_{OUT}}{\Delta I_L} = \frac{r_{ds} || R_L}{1 + A_{EA} A_{MP}} \approx \frac{R_L}{A_{OL}} \quad (6)$$

1.2.4 Load Transient Response Time (TR) & Settling Time (TS):

As mentioned in Section 1.1.1, if the I_L changes from light to heavy, C_L initially compensates the current, and V_{OUT} falls. After that, feedback loop operates with the speed of loop bandwidth and since I_{OUT} at that time can provide same quantity of current with I_L , the V_{OUT} can be same with V_{REF} .

T_R is defined as the time for the changed V_{OUT} to be compensated and head toward the direction of the desired value. In Figure 1.10, (7), and (8), the calculation of T_R is introduced according to the relationship between T_R and I_L transition time (T_{edge}) where t_0 is the moment at which the I_L changes. T_S is defined as the time that V_{OUT} that deviates from V_{REF} returns back near V_{REF} , typically entering the range of 2% error with V_{REF} . In Figure 1.11, the metrics are marked on the graph of V_{OUT} transition.

$$\begin{aligned} \Delta V_{OUT} &= \frac{1}{C_L} \left(\int_{t_0}^{t_0 + T_{edge}} \Delta I_{OUT}(t) dt + \int_{t_0 + T_{edge}}^{t_0 + T_R} \Delta I_{OUT} dt \right) \\ &\approx \frac{\Delta I_{OUT} T_R}{C_L} \end{aligned} \quad (7)$$

Figure 1.10 TR when (a) $T_{edge} \ll T_R$ and (b) $T_{edge} > T_R$

$$\begin{aligned}\Delta V_{OUT} &= \frac{1}{C_L} \int_{t_0}^{t_0+T_R} \Delta I_{OUT}(t) dt, \\ \Delta V_{OUT} &= \frac{T_R \Delta I_{OUT}(t)}{2C_L} \Big|_{t=T_R}, \\ \Delta I_{OUT}(t) \Big|_{t=T_{edge}} &= \Delta I_{OUT} \rightarrow \frac{dI_{OUT}}{dt} = \frac{\Delta I_{OUT}}{T_{edge}} \rightarrow \Delta I_{OUT}(t) \Big|_{t=T_R} = \frac{\Delta I_{OUT} T_R}{T_{edge}}, \\ T_R &= \sqrt{\frac{2C_L \Delta V_{OUT} T_{edge}}{\Delta I_{OUT}}}\end{aligned}\tag{8}$$

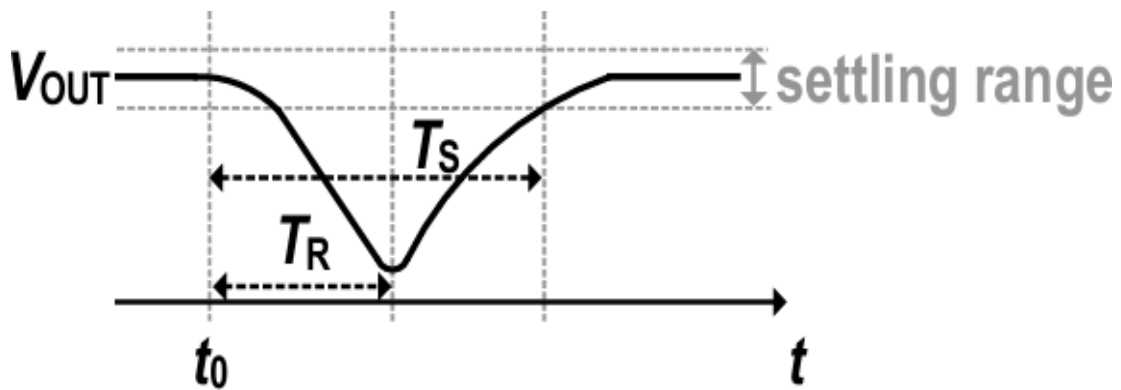


Figure 1.11 TR and TS when load transient.

1.2.5 Dropout Voltage (VDO):

Quiescent current (I_Q) is defined as the difference between input and output currents of LDO. I_Q is shown in Figure 1.12 and (9).

$$I_Q = I_{IN} - I_{OUT} \quad (9)$$

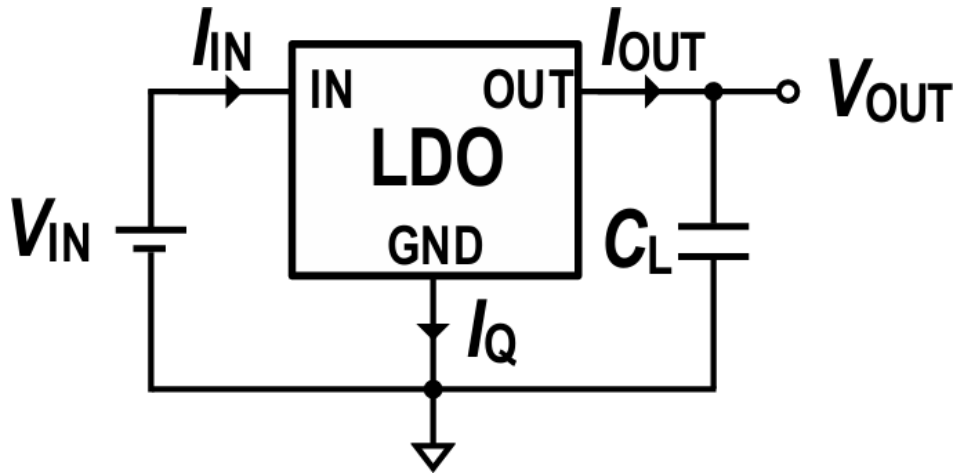


Figure 1.12 I_Q in LDO.

VDO and I_Q are the metrics related to the power of LDO. These metrics are important for battery based applications since power reduction can prolong the battery life. The power efficiency of LDO can be written as

$$\begin{aligned} \text{Power efficiency} &= \frac{I_{OUT} V_{OUT}}{(I_{OUT} + I_Q) V_{IN}} \times 100 \\ &= \eta_I \left(1 - \frac{V_{DO}}{V_{IN}}\right) \times 100 \end{aligned} \quad (10)$$

where η_I is the current efficiency. The focus whether reducing VDO or reducing I_Q can be judged by the load condition. If the $I_{L,max}$ is huge and LDO usually operate in heavy load condition, reducing VDO is more effective for reducing power. However, if the load operates usually in light load condition, the η_I will affect overall power significantly, so I_Q will be more important. It depends on how long the LDO stays in ON state and how much I_L is needed averagely. Designers can observe the load characteristics carefully and adopt right strategies.

1.3 Motivation:

The motivation for this project arises from the rapid growth of wearable health monitoring devices, such as smartwatches and biosensors, which demand highly efficient, compact, and reliable power management solutions. These devices operate on limited battery resources and often include sensitive analog and mixed-signal circuits that process low-amplitude biological signals. Even minor voltage fluctuations or noise in the power supply can significantly degrade the accuracy of measurements like ECG, EEG, and heart rate signals. Conventional voltage regulators struggle to meet these stringent requirements due to limitations in Power Supply Rejection Ratio (PSRR), line and load regulation, and transient response. Hence, there is a strong need to design an advanced Low Dropout Regulator (LDO) that can deliver a stable, low-noise output while operating efficiently under varying input and load conditions.

This project is motivated by the need to enhance LDO performance by improving key parameters such as PSRR, stability, and regulation accuracy, specifically for wearable health applications. By leveraging Cadence-based design and simulation, the aim is to develop an LDO that not only maintains a consistent output voltage but also effectively suppresses supply noise across a wide frequency range. Additionally, achieving low dropout voltage, fast transient response, and robust operation under process, voltage, and temperature (PVT) variations is critical for real-world deployment. The proposed design addresses these challenges by optimizing the error amplifier, compensation network, and feedback mechanism, ultimately contributing to the development of reliable, low-power, and high-performance power management systems for next-generation wearable electronics.

1.4 Problem Statement and Objective:

Problem Statement:

- Wearable power management units require a high-PSRR, low-dropout and stable output under varying load and supply conditions. However, achieving these performance targets simultaneously while maintaining post-layout stability and robustness across PVT variations presents significant design challenges due to trade-offs between noise rejection, power efficiency, and loop stability.

Objective:

- To design a high-PSRR, low-dropout (LDO) regulator for wearable applications with stable output across PVT variations. (Phase 1)
- To implement the physical layout of the LDO and verify its performance through post-layout simulations. (Phase 2)

1.5 Scope of this project:

- This project can be further optimized for ultra-low power consumption to extend battery life in next-generation wearable and implantable medical devices
- The scope to integrate the LDO into full System-on-Chip (SoC) designs with advanced compensation techniques for improved miniaturization
- Also can be enhanced using adaptive or digital-assisted control methods to achieve higher efficiency and better dynamic performance under varying loads
- Potential to extend the design for high-frequency and RF applications requiring improved PSRR and noise immunity in advanced communication systems.

CHAPTER 2: Literature Survey

2.1 High-PSRR LDO with Low Noise and Ultra-Low Power Consumption:

- Guo et al. presented an LDO targeting simultaneous high PSRR, low output noise and ultra-low standby current. The paper highlights that many prior LDOs improve one metric but fail to meet PSRR, noise and standby current requirements at the same time. Their approach increases loop gain, introduces an improved feed-forward path and uses isolated supply techniques, reaching up to 95 dB PSRR at 1 kHz and high-frequency PSRR above 45 dB [1].
- This work is important for the present project because it shows that high PSRR is achieved not by a single device sizing decision but through a combination of loop gain, ripple cancellation and noise filtering. The proposed project follows the same design philosophy at the academic simulation level by emphasizing PSRR, line/load regulation and stability rather than reporting output voltage alone [1], [5].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.2 FVF-CFFRC Hybrid LDO for High PSR and Fast Transient Response:

- Yang et al. proposed a flipped-voltage-follower/current-mode feed-forward ripple cancellation hybrid LDO. Their design uses current-mode feed-forward ripple cancellation to improve PSR while preserving the fast response advantage of the FVF architecture. The paper reports about 30 dB PSR improvement at 1 kHz and fast recovery during current steps [2].
- This paper is relevant because it links PSRR and transient response, two parameters that are often treated separately. The proposed project also analyzes both PSRR and transient response; the Cadence results show high low-frequency rejection and fast settling behavior, which are essential for load-varying wearable circuits [2], [4].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.3 FVF-Based LDO With Scaled-Down Current Buffer Compensation

- Qin, Ki and Law studied an FVF-based LDO with scaled-down current buffer compensation for a wide load current range. The work emphasizes that stability becomes challenging when the output capacitor and load current vary, and it uses compensation and transient current enhancement to reduce output spikes [3].
- The present project uses a compensation network containing R_c and C_c , and the loop gain plot confirms a stable phase margin. The comparison with this paper helps justify the inclusion of stability analysis as a separate report section rather than only showing DC output regulation [3], [10].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.4 High-Efficiency LDO With Adaptive PSRR and Bandwidth

Enhancements:

- Zeznick et al. introduced adaptive gain boosting and adaptive Miller compensation for a high-efficiency output-capacitor-less LDO. Their idea is to increase PSRR and bandwidth when the load current requires it, while preserving efficiency at lighter loads. The paper is explicitly connected to biosensing and IoT applications, where supply noise corrupts low-amplitude biological signals [4].
- This paper closely matches the application context of the present report. It supports the argument that high PSRR is critical in wearable health monitoring devices and that the LDO should be evaluated over frequency and load conditions. The project PSRR of approximately -80 dB in the lower-to-mid frequency region is therefore a central result [4], [5].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.5 Low Dropout, High PSRR LDO for Wearable Devices:

- Adil et al. presented a Cadence Virtuoso LDO for wearable device PMUs using 180 nm GPDK CMOS. Their design regulates 1.64 V from 1.8 V, reports 160 mV dropout, gain around 77.58 dB, phase margin around 65 degrees and PSRR around -60 dB at kHz/MHz ranges [5].
- This paper is the closest direct comparison for the project because both works use Cadence and target wearable applications. The proposed project obtains $V_{out} \approx 1.702$ V from $V_{in} = 1.8$ V, dropout ≈ 98 mV,

phase margin ≈ 66.86 degrees and stronger PSRR around -80 dB at low and mid frequencies, showing improvement in key regulation and rejection metrics [5].

- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.6 Design and Performance Analysis of LDO With High PSRR:

- Patgar et al. analyzed a high-PSRR LDO using a two-stage OTA error amplifier and NMOS pass element. The paper reports parameters including 1.8 V input, 0.8 V output, PSRR around -50 dB at 100 kHz, loop gain around 60 dB and phase margin around 60 degrees [6].
- The present design uses similar evaluation metrics but targets a different output voltage and application space. Its line regulation, load regulation and phase margin are reported in a results table, which allows a direct comparison against the reported loop gain and stability values in this paper [6].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.7 0.9-uA Quiescent Current High-PSRR LDO Using CFFRC:

- Guo, Kang and Roh proposed a low-quiescent-current high-PSRR LDO using capacitive feed-forward ripple cancellation. The design consumes only 0.9 uA quiescent current and improves measured PSRR by about 22 dB at 1 MHz at maximum load current. The paper explains that supply noise couples through the error amplifier, pass device capacitances and pass transistor output impedance [7].
- This work is used in the report to explain the physical paths by which power-supply ripple reaches V_{out} . Even though the project does not implement the same CFFRC architecture, the PSRR section uses this theory to interpret why strong loop gain and compensation help reduce ripple transmission [7].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.8 High-PSRR NMOS LDO With Intrinsic Gain-Tracking Ripple Cancellation:

- Kim et al. proposed an NMOS LDO using intrinsic gain-tracking ripple cancellation and adaptive biasing. The paper reports low quiescent current and high PSRR over a wide frequency range, with careful attention to how the pass device and replica structures track variations [8].
- The present project is compared with this work because the measured PSRR behavior and load regulation depend on the pass transistor operating region. The uploaded PVT and load plots are interpreted with the same idea that process and bias variations change gain, output impedance and pole locations [8], [12].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.9 PSRR-Enhanced Fast-Response Inverter-Based LDO:

- Lee, Lim and Han proposed a low-voltage inverter-based LDO for mobile devices. The paper addresses the difficulty of obtaining high gain under low supply voltage and reports fast settling times together with PSRR at 100 kHz and 10 MHz [9].
- This paper is useful because it connects low-voltage operation with fast transient response. The project also works with a 1.8 V input and shows a 5 μ s settling time with small overshoot and undershoot, which makes the transient behavior acceptable for wearable load transitions [9].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.10 Stability-Enhanced LDO With High PSRR in 40-nm CMOS:

- Gorospe et al. presented a stability-enhanced LDO with high PSRR for SoC applications. The paper specifically emphasizes the trade-off between high PSRR and robust stability at large load currents, and introduces adaptive zero compensation and dc-coupled zero compensation to maintain phase and gain margins [10].
- This paper strongly supports the separate stability section of the report. The project reports a phase margin around 66.86 degrees and gain margin around 79.02 dB, both of which exceed conventional minimum stability expectations and show that the compensation network produces a stable loop [10].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.11 Fully Integrated 180 nm CMOS 1.2 V LDO for Portable Applications:

- Perez-Bailon, Calvo and Medrano presented a fully integrated capacitor-less 180 nm CMOS LDO for low-power portable applications. Their paper highlights the need for low quiescent current, small area and stable operation over battery voltage and temperature range [11].
- The present report uses this paper to discuss integration and wearable suitability. The proposed design is not claimed as a fabricated chip, but its Cadence verification of DC, AC, PVT and transient behavior follows the same design logic needed before layout and integration [11].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.12 LDO PSpice Modeling With Dependency on ESR, Cout and Temperature:

- Rao and Pai provided a tutorial view of LDO modeling and how dropout, stability, load transient response, PSRR, Cout, ESR and temperature are interconnected. The paper notes that PSRR and stability can change with temperature and load because loop gain, output impedance and pole positions change [12].
- This work is important for interpreting the PVT plots in the project. The uploaded PVT analysis across TT, FF, SS, FS and SF corners at different voltages and temperatures shows that corner verification is necessary because nominal stability alone is not enough [12].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

2.13 Fast-Transient LDO With Pseudo-ESR Technique:

- Li et al. proposed a 600 mA fast-transient LDO using dual-loop compensation and pseudo-ESR technique. Their work shows that load transients can cause output overshoot and undershoot and that compensation must be designed to reduce the output spike while preserving loop bandwidth [13].
- The project transient response section uses this idea to interpret the observed 0.5 mV undershoot/overshoot and 5 μ s settling time. Although the project load current is smaller, the same control principle applies: the LDO must quickly correct Vout after sudden load disturbance [13].
- The key lesson taken from this paper is integrated into the proposed design evaluation: an LDO report should not stop at schematic design but must quantify dropout, PSRR, line regulation, load regulation, stability and transient response using clearly stated test conditions [1]-[13].

CHAPTER 3: Proposed LDO Design Objective

- The objective of the proposed design is to create a Cadence-simulated LDO with low dropout, high PSRR, improved line regulation and improved load regulation for wearable health monitoring systems. The circuit is evaluated not only from a DC operating point but also from PSRR, loop stability, PVT and transient response results [4], [5], [10].
- The design target is based on the practical observation that wearable devices require quiet supply rails while operating from small batteries. A regulator with a 98 mV dropout can continue regulating even when the battery voltage approaches the required output level, while high PSRR suppresses ripple from upstream converters or digital switching activity [4], [7].
- The final result table reports 90 nm technology, $V_{in} = 1.8$ V, $V_{out} = 1.702$ V, dropout = 98 mV, line regulation = 0.146 mV/V, load regulation = 1.26 mV/mA, phase margin = 66.6561 degrees, gain margin = 79.0029 dB, 0.5 mV undershoot/overshoot and 5 μ s settling time [5], [10].

3.1 Proposed Methodology:

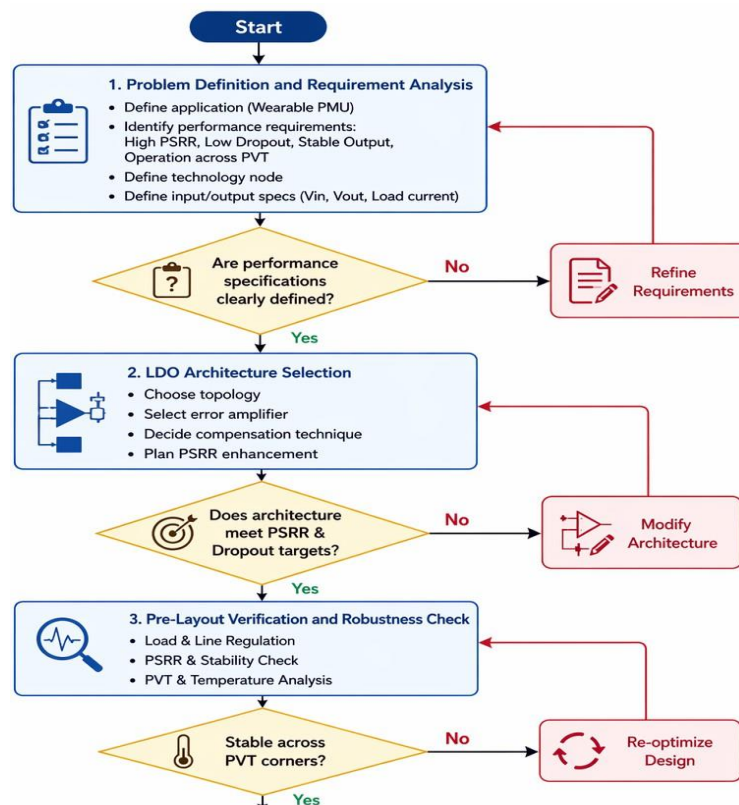


Figure 3.1 Proposed Methodology

3.2 Block Diagram and Circuit-Level Architecture:

- As shown in Figure 3.2, The block diagram contains the same functional elements used in classical LDOs: an error amplifier, pass transistor, compensation network, feedback divider, output capacitor with ESR and load current source. V_{ref} is compared with V_{fed} , and the amplifier output controls the pass transistor so that the output remains regulated [5], [6].
- The Cadence transistor-level schematic expands the block diagram into MOS devices. The current bias network supplies the error amplifier, the differential pair compares V_{ref} and V_{fed} , and PMOS devices form the pass path and active loads. R_c and C_c are introduced for compensation to improve phase margin and reduce instability risk [10], [12].
- This architecture is intentionally practical for academic implementation because it allows each major metric to be simulated directly: DC operating point for dropout, DC sweep for line/load regulation, AC analysis for PSRR and loop gain, transient analysis for settling, and corner sweeps for PVT robustness [5], [12].

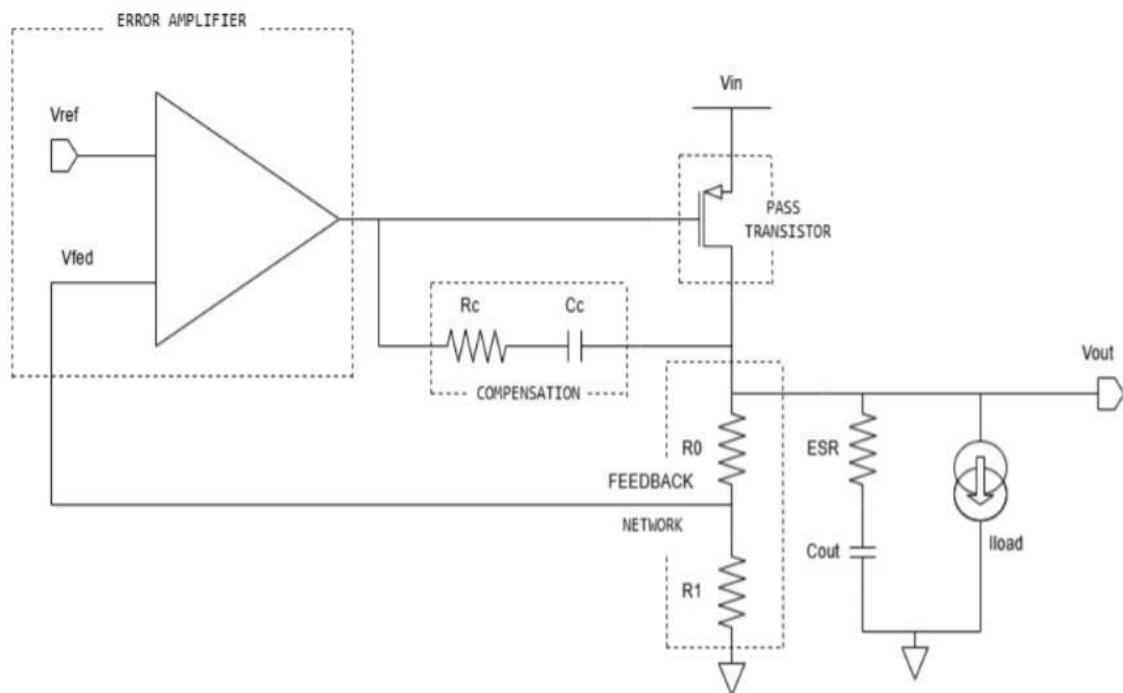


Figure 3.2 Functional block diagram of the proposed LDO.

3.2.1 Working:

I. Error amplifier

Function:

The error amplifier acts as the brain of the LDO. It constantly checks whether the output voltage is at the desired level.

Working:

It takes two inputs: the reference voltage and the feedback voltage. The feedback voltage is a scaled version of the output. The amplifier compares these two values and produces an output signal based on the difference between them. If the output voltage drops below the required level, the error amplifier increases its output to correct it. If the output voltage rises too much, it reduces the signal. This continuous adjustment helps maintain a steady output voltage under all conditions.

II. Reference voltage (V_{ref})

Function:

The reference voltage provides a stable and fixed value that the output voltage tries to follow.

Working:

This voltage is designed to remain constant regardless of changes in temperature or input supply. It serves as a benchmark for the entire regulation process. The error amplifier always tries to make the feedback voltage equal to this reference value, which indirectly controls the final output voltage.

III. Pass transistor

Function:

The pass transistor controls how much current flows from the input to the output.

Working:

It behaves like a controllable resistor. The gate of the transistor is driven by the error amplifier. When more current is needed at the output, the transistor allows more current to pass through. When less current is required, it restricts the flow. This adjustment directly controls the output voltage. The performance of this transistor also affects important parameters like dropout voltage and efficiency.

IV. Compensation network (R_c and C_c)

Function:

The compensation network ensures that the system remains stable and does not oscillate.

Working:

Since the LDO uses a feedback loop, there is a possibility of instability if the loop is not properly controlled. The RC network introduces specific frequency characteristics that improve the phase

margin. This helps the system respond smoothly to changes without causing unwanted oscillations or ringing in the output voltage.

V. Feedback network (R0 and R1)

Function:

The feedback network samples the output voltage and sends a portion of it back to the error amplifier.

Working:

The resistors form a voltage divider. Instead of feeding the full output voltage, only a scaled-down version is sent back. This makes it easier to compare with the reference voltage. By choosing proper resistor values, the desired output voltage can be set. This network is essential for maintaining accurate regulation.

VI. Output capacitor and ESR

Function:

The output capacitor helps in smoothing the output voltage and improving stability.

Working:

The capacitor stores energy and releases it when there is a sudden increase in load demand. This reduces voltage dips. It also filters out high-frequency noise, making the output cleaner. The ESR of the capacitor plays a helpful role by introducing a zero in the system, which improves stability and transient response.

VII. Load

Function:

The load represents the actual circuit or device powered by the LDO.

Working:

In real applications, the load current is not constant and can change frequently. When the load increases, the output voltage tends to drop, and when the load decreases, the voltage may rise. The LDO reacts quickly to these changes by adjusting the pass transistor, ensuring that the output voltage remains stable.

3.3 Cadence Transistor-Level Schematic:

Figure 3.3 shows the schematic of the current bias source, NMOS differential input pair, current mirror loads, compensation network and PMOS pass path. The feedback node V_{fed} is connected from the output through the resistor divider, enabling closed-loop regulation [5], [6].

The compensation resistor R_c and capacitor C_c are placed to shape the loop response. Without proper compensation, the output pole, pass-transistor gate pole and amplifier internal poles can reduce phase margin and cause ringing or oscillation. The stability plot later confirms that this network provides a safe phase margin [10], [12].

The schematic therefore supports the report claim that the design is not only a voltage divider or open-loop circuit, but a closed-loop analog regulator with feedback, biasing, pass control and frequency compensation [5], [10].

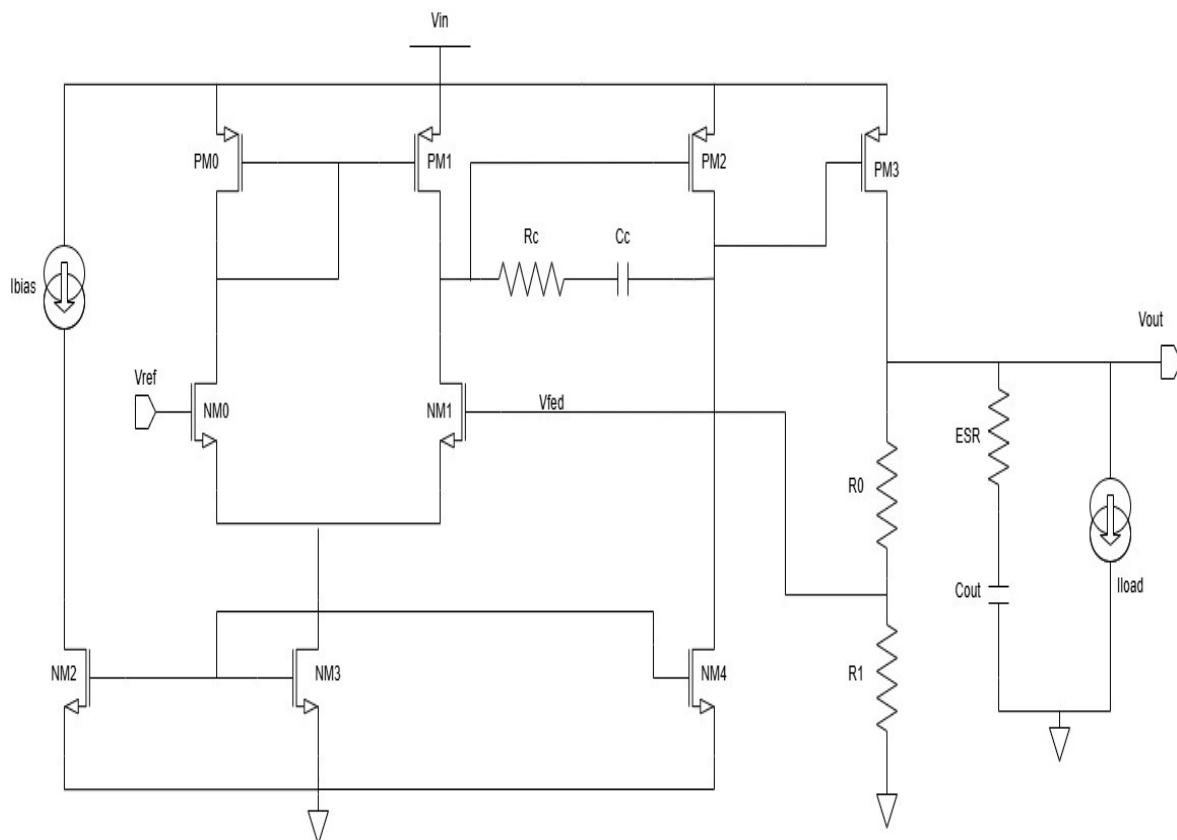


Figure 3.3 Cadence transistor-level schematic of the LDO.

3.3.1 Components and its functions:

- Bias Current Source (I_{bias}):
 - Provides a constant current to the internal circuit blocks.
 - Ensures proper biasing of transistors (NMOS & PMOS) in the error amplifier.
 - Helps maintain stable operation independent of supply variations.
- Error Amplifier (PM0, PM1, NM0, NM1):
 - Acts as the control unit of the LDO.
 - Compares : Reference voltage (V_{ref}) and Feedback voltage (V_{fed})
 - Generates an error signal that controls the pass transistor.
 - High gain ensures accurate output voltage regulation.
- Pass Transistor (PM2, PM3):
 - Main power control element of the LDO.
 - Works like a variable resistor between V_{in} and V_{out}.
 - Controlled by the error amplifier output.
 - Regulates how much current flows to the load.
 - Compares : Dropout voltage and Efficiency
- Compensation Network (R_c, C_c):
 - Used for stability of the feedback loop
 - Introduces poles and zeros to control frequency response.
 - Prevents oscillations in output voltage.
- Feedback Network (R₀, R₁):
 - Forms a voltage divider from V_{out}.
 - Generates feedback voltage:

$$V_{fed} = V_{out} \times \frac{R_1}{R_0 + R_1}$$

- Ensures output voltage is maintained at:

$$V_{out} = V_{ref} \left(1 + \frac{R_0}{R_1}\right)$$

- Enables closed-loop regulation.
- Output Capacitor (C_{out}) and ESR:

- Enhance transient response
- Stabilize the LDO
- Load (I_{load}):
 - Represents the actual device consuming power.
 - Load current variations test: Load regulation and transient response.
 - LDO adjusts automatically to maintain constant V_{out} .

3.4 Cadence Implementation Methodology

The Cadence methodology starts with drawing the schematic and defining the input, output, reference, feedback and load nodes. Device sizes are selected so that the pass transistor can supply the load current while maintaining dropout and stability. The bias current is chosen to keep the error amplifier active without excessive power consumption [5], [11].

DC simulation is used first because it confirms that the circuit reaches the intended operating point. After the DC operating point is validated, line and load sweeps are performed to determine how sensitive the output voltage is to V_{in} and I_{load} . AC analysis is then used for PSRR and loop gain, while transient analysis validates time-domain response [12], [13].

This Figure 3.4 shows sequence follows a typical analog design verification flow: first verify biasing, then small-signal behavior, then dynamic behavior, and finally corner behavior. The report uses attached simulation images from each stage so that the result interpretation is traceable to the Cadence outputs [5], [12].

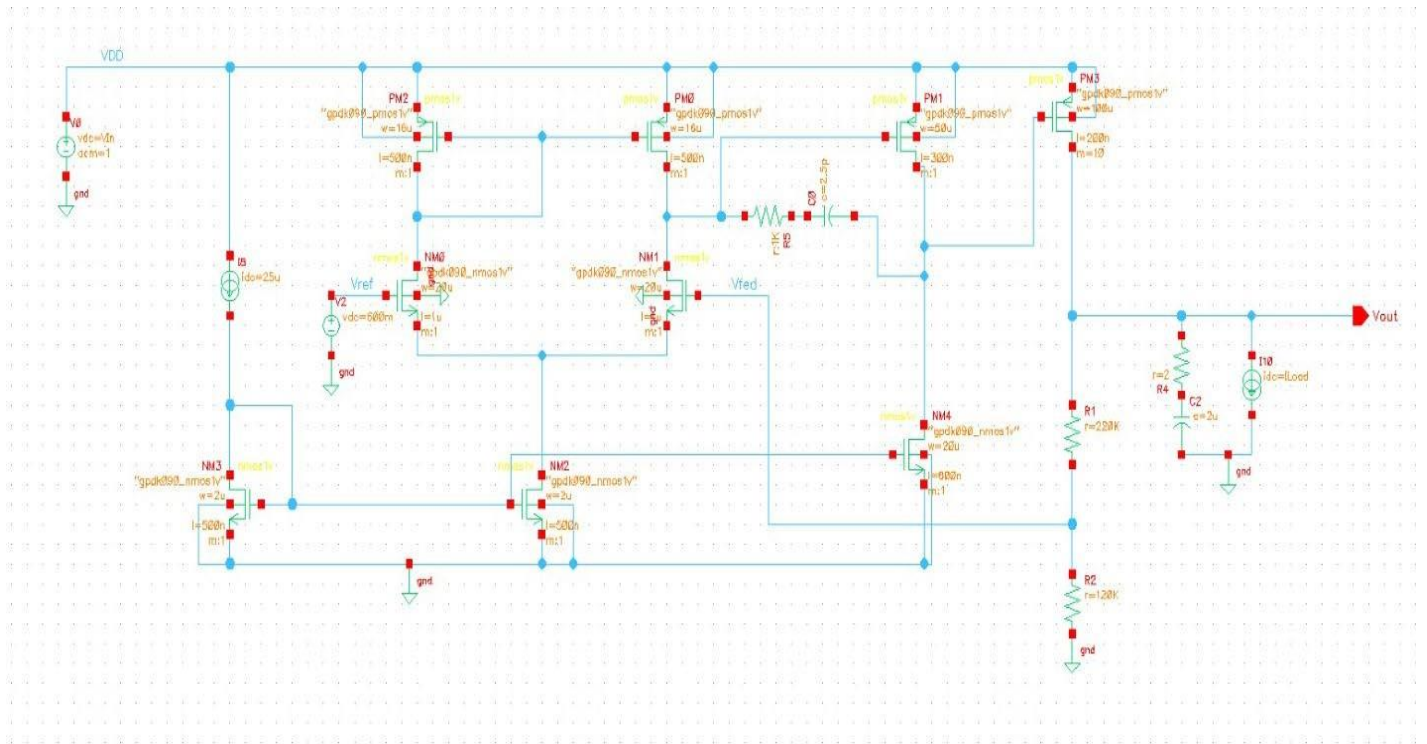


Figure 3.4 Schematic in Cadence Virtuoso

3.4.1 Working of Proposed LDO at Circuit Level:

1. Circuit idea

- This schematic represents the transistor-level implementation of the proposed low dropout regulator.
- The main purpose of the circuit is to convert an input supply voltage into a stable regulated output voltage.
- In this design, the LDO takes input from the VDD/V_{in} rail and provides a regulated output at V_{out}.
- The circuit is made up of a biasing section, error amplifier, pass transistor stage, compensation network, feedback network, output capacitor, ESR path, and load current source.
- All these parts work together as a closed-loop system to maintain a nearly constant output voltage even when the input voltage or load current changes.
- The regulator continuously senses the output voltage, compares it with a reference voltage, and adjusts the pass transistor to correct any error.

2. Input supply and ground

- The top horizontal rail in the schematic represents the supply voltage line, marked as VDD or V_{in}.
- This supply provides power to the complete LDO circuit.
- The ground nodes are connected at different places in the circuit and act as the common reference point for all voltages.
- A stable input supply is important, but even if the input has small variations, the LDO is designed to reduce their effect at the output.
- The difference between V_{in} and V_{out} is the dropout voltage, which determines how close the output voltage can remain to the input voltage while still maintaining regulation.

3. Bias current source

- The current source on the left side provides the bias current required for the internal transistor stages.
- This bias current sets the operating point of the amplifier and current mirror circuits.
- Without proper biasing, the MOS transistors would not operate in the correct region, and the LDO would not regulate properly.

- The bias current also influences power consumption, speed, and stability.
- A lower bias current helps reduce power consumption, which is useful for wearable devices, but it may reduce the speed of response.
- A properly chosen bias current gives a balance between low power operation and good transient performance.

4. Current mirror section

- The transistors near the left and bottom side of the schematic form current mirror branches.
- Current mirrors are used to copy or replicate the reference bias current into different parts of the circuit.
- They ensure that the differential amplifier and other internal stages receive suitable current.
- The current mirror improves matching and allows the circuit to operate consistently.
- In this design, the NMOS transistors at the bottom help sink current from the amplifier branches.
- This makes the amplifier operate properly and supports stable comparison between V_{ref} and V_{fed} .

5. Error amplifier section

- The central part of the schematic contains the error amplifier.
- The error amplifier is one of the most important parts of the LDO because it controls the regulation action.
- It compares two voltages: the reference voltage V_{ref} and the feedback voltage V_{fed} .
- V_{ref} is the fixed target voltage, while V_{fed} is taken from the output through the resistor feedback network.
- If V_{fed} is lower than V_{ref} , it means the output voltage has dropped.
- The error amplifier responds by changing its output so that the pass transistor supplies more current.
- If V_{fed} is higher than V_{ref} , it means the output voltage has increased.
- The amplifier then adjusts the pass transistor in the opposite direction to reduce the output.
- This continuous comparison and correction is what keeps the LDO output stable.

6. Differential input pair

- The two transistors connected to V_{ref} and V_{fed} act like a differential input pair.
- One transistor senses the reference voltage, and the other senses the feedback voltage.
- The pair converts the difference between these two voltages into a current or voltage change inside the circuit.
- This difference is very small during normal operation because the feedback loop tries to make V_{fed} nearly equal to V_{ref} .
- A well-designed differential pair improves accuracy and helps achieve good line and load regulation.
- In this project, the differential input pair helps the LDO maintain an output voltage close to the expected value.

7. Active load transistors

- The PMOS transistors connected above the differential pair act as active loads.
- Active loads are used instead of simple resistors because they provide higher gain in CMOS circuits.
- Higher gain is important because it allows the error amplifier to detect even small changes in output voltage.
- This helps improve regulation accuracy.
- The active load also helps convert changes in current from the differential pair into a useful voltage signal.
- This signal is then used to control the next stage of the LDO.

8. Gain stage and control node

- The output of the error amplifier is connected toward the gate control path of the pass transistor.
- This node determines how strongly the pass transistor is turned on or off.
- When output voltage correction is needed, the control node voltage changes accordingly.
- This node is important because it links the sensing part of the LDO to the power delivery part.
- A strong and stable control signal ensures that the pass device responds correctly.
- If this node is poorly controlled, the output may oscillate or show large overshoot and undershoot.

9. Pass transistor section

- The PMOS transistor on the right side acts as the main pass transistor.
- It is placed between the input supply and the output node.
- Its function is to deliver current from V_{in} to V_{out} .
- The pass transistor behaves like a variable resistor controlled by the error amplifier.
- When the load needs more current, the error amplifier drives the gate so that the pass transistor conducts more.
- When the load current decreases, the pass transistor conducts less.
- This action keeps the output voltage close to the desired regulated value.
- The size of the pass transistor strongly affects dropout voltage and load current capability.
- A larger pass transistor can supply more current with less voltage drop, but it also increases capacitance and may affect stability.

10. Dropout behavior

- The dropout voltage is the minimum voltage difference required between input and output for the LDO to remain in regulation.
- In this project, the input is around 1.8 V and the output is around 1.702 V.
- This gives a dropout voltage of approximately 98 mV.
- A low dropout voltage is useful because it allows the circuit to operate efficiently even when the battery voltage decreases.
- This is especially important for wearable devices, where battery life is a major concern.
- The pass transistor plays a major role in achieving this low dropout operation.

11. Compensation network

- The resistor and capacitor connected near the control path form the compensation network.
- This network usually contains R_c and C_c .
- Its main purpose is to improve the stability of the LDO feedback loop.
- Since the LDO uses feedback, there is always a possibility of oscillation if the poles and zeros are not properly controlled.

- The compensation network shapes the frequency response of the loop.
- It helps improve phase margin and allows the output to settle smoothly after sudden changes.
- In this project, the compensation network supports the achieved phase margin and stable transient behavior.

12. Feedback network

- The two resistors connected from the output node to ground form the feedback network.
- This network samples the output voltage and produces the feedback voltage V_{fed} .
- The feedback voltage is sent back to the error amplifier.
- The resistor ratio decides the final output voltage of the LDO.
- The basic relationship is:

$$V_{fed} = V_{out} \times R_{lower} / (R_{upper} + R_{lower})$$

- During regulation, the error amplifier tries to make V_{fed} equal to V_{ref} .
- Therefore, the output voltage depends on the reference voltage and feedback resistor ratio.
- This network is essential because it closes the loop and allows automatic correction.

13. Output capacitor and ESR

- The output side contains a capacitor connected to ground along with an ESR path.
- The capacitor helps smooth the output voltage.
- When the load current changes suddenly, the capacitor temporarily supplies or absorbs charge.
- This reduces sudden voltage dips or spikes.
- ESR stands for equivalent series resistance.
- In an LDO, ESR can also help with stability by introducing a zero in the frequency response.
- The combination of output capacitor and ESR improves transient response and reduces ripple.

14. Load current source

- The current source connected at the output represents the load current.

- It models the current drawn by the external circuit powered by the LDO.
- In actual wearable or portable devices, the load current changes when different blocks turn on or off.
- For example, sensors, signal processors, or wireless modules may draw different currents at different times.
- The LDO must respond to these changes and keep V_{out} stable.
- The load current source in the simulation is used to test load regulation and transient behavior.

15. Output node

- The output node is marked as V_{out} on the right side of the schematic.
- This is the regulated voltage delivered to the load.
- The goal of the entire circuit is to keep this node stable.
- In your simulation, V_{out} is around 1.702 V.
- A stable V_{out} indicates that the error amplifier, pass transistor, compensation network, and feedback network are working correctly.
- Since wearable circuits need clean power, the quality of this output is very important.

16. Working during normal operation

- The circuit starts with the bias current source setting the operating current of the internal transistors.
- The reference voltage is applied to one input of the error amplifier.
- A portion of the output voltage is fed back to the other input through the feedback resistors.
- The error amplifier compares V_{ref} and V_{fed} .
- If both voltages are nearly equal, the circuit is in regulation.
- The pass transistor then supplies just enough current to maintain the output voltage.
- The output capacitor helps keep the output smooth.
- The compensation network ensures that the correction does not become unstable.

17. Working when output voltage decreases

- If the load current increases, the output voltage tends to decrease.

- This causes the feedback voltage V_{fed} to decrease.
- The error amplifier detects this difference between V_{ref} and V_{fed} .
- It changes the gate voltage of the pass transistor so that more current flows from V_{in} to V_{out} .
- The output capacitor also supports the load temporarily during this sudden change.
- As more current is supplied, V_{out} returns to its regulated value.
- This explains the load regulation and transient response behavior of the LDO.

18. Working when output voltage increases

- If the load current suddenly decreases, the output voltage may rise slightly.
- This increase also raises the feedback voltage.
- The error amplifier senses that V_{fed} is higher than the reference condition.
- It reduces the pass transistor conduction.
- As a result, less current flows to the output.
- The output voltage then settles back to the desired value.
- This correction prevents excessive overshoot and keeps the output safe for the load circuit.

19. Relevance to line regulation

- Line regulation depends on how well the LDO maintains V_{out} when V_{in} changes.
- In this schematic, variations in V_{in} are mainly rejected through the combined action of the error amplifier and pass transistor.
- If V_{in} changes, the output may try to shift.
- The feedback loop senses that shift and corrects the pass transistor drive.
- Good loop gain helps reduce output variation.
- This is why the schematic supports improved line regulation performance.

20. Relevance to load regulation

- Load regulation depends on how well the LDO maintains V_{out} when I_{load} changes.

- In this schematic, the pass transistor supplies the required load current.
- The feedback network senses output movement caused by load changes.
- The error amplifier adjusts the pass transistor accordingly.
- The output capacitor reduces sudden voltage movement.
- Together, these blocks help achieve good load regulation.

21. Relevance to PSRR

- PSRR shows how well the LDO rejects noise from the input supply.
- In this schematic, input noise can try to pass through the supply line and pass transistor to the output.
- The error amplifier and feedback loop help cancel this disturbance.
- The compensation network and output capacitor also help reduce noise at different frequencies.
- A stronger error amplifier and properly designed pass transistor improve PSRR.
- This is why the proposed LDO is suitable for low-noise applications.

22. Relevance to stability

- Since this LDO works using feedback, stability is very important.
- The schematic includes compensation components to control the loop response.
- The output capacitor and ESR also contribute to stability.
- The loop must respond fast enough to correct errors, but not so aggressively that it oscillates.
- The achieved phase margin in the project shows that the circuit has stable operation.

23. Relevance to wearable devices

- Wearable health monitoring devices need stable and clean power because they measure weak biological signals.
- Noise or voltage fluctuation can affect sensors and analog front-end circuits.
- This LDO schematic is designed to provide low-noise regulated voltage.
- The low dropout voltage supports longer battery usage.

- Good regulation and PSRR make it useful for health monitoring systems such as ECG, heart-rate, and portable biosensor circuits.

24. Final understanding

- This schematic shows a complete LDO system at transistor level.
- The bias circuit provides operating current.
- The error amplifier detects output errors.
- The pass transistor controls power delivery.
- The feedback network sets and monitors output voltage.
- The compensation network improves stability.
- The output capacitor and ESR support smooth and stable output.
- The load current source represents real circuit demand.
- Together, these blocks allow the LDO to maintain a stable output voltage with good line regulation, load regulation, PSRR, and transient response.

CHAPTER 4: Simulation Results

4.1 DC Operating Point and Dropout Analysis:

As shown in Figure 4.1, The DC operating point confirms whether the regulator can produce a stable output voltage under nominal input supply. The attached dropout plot shows V_{in} fixed at about 1.8 V and V_{out} stabilized near 1.701-1.702 V. This confirms that the feedback loop has settled and the pass transistor remains in regulation [5], [6].

Dropout voltage is calculated using $V_{dropout} = V_{in} - V_{out}$. With $V_{in} = 1.8$ V and $V_{out} = 1.702$ V, the dropout is 0.098 V or 98 mV. A small dropout voltage is highly useful for wearable devices because it allows the battery to discharge closer to the desired rail while still maintaining regulation [5], [11].

Compared with the wearable LDO paper that reports 160 mV dropout from 1.8 V to 1.64 V, the present project shows an improved nominal dropout of about 98 mV. This improves headroom utilization and supports low-voltage battery operation [5].

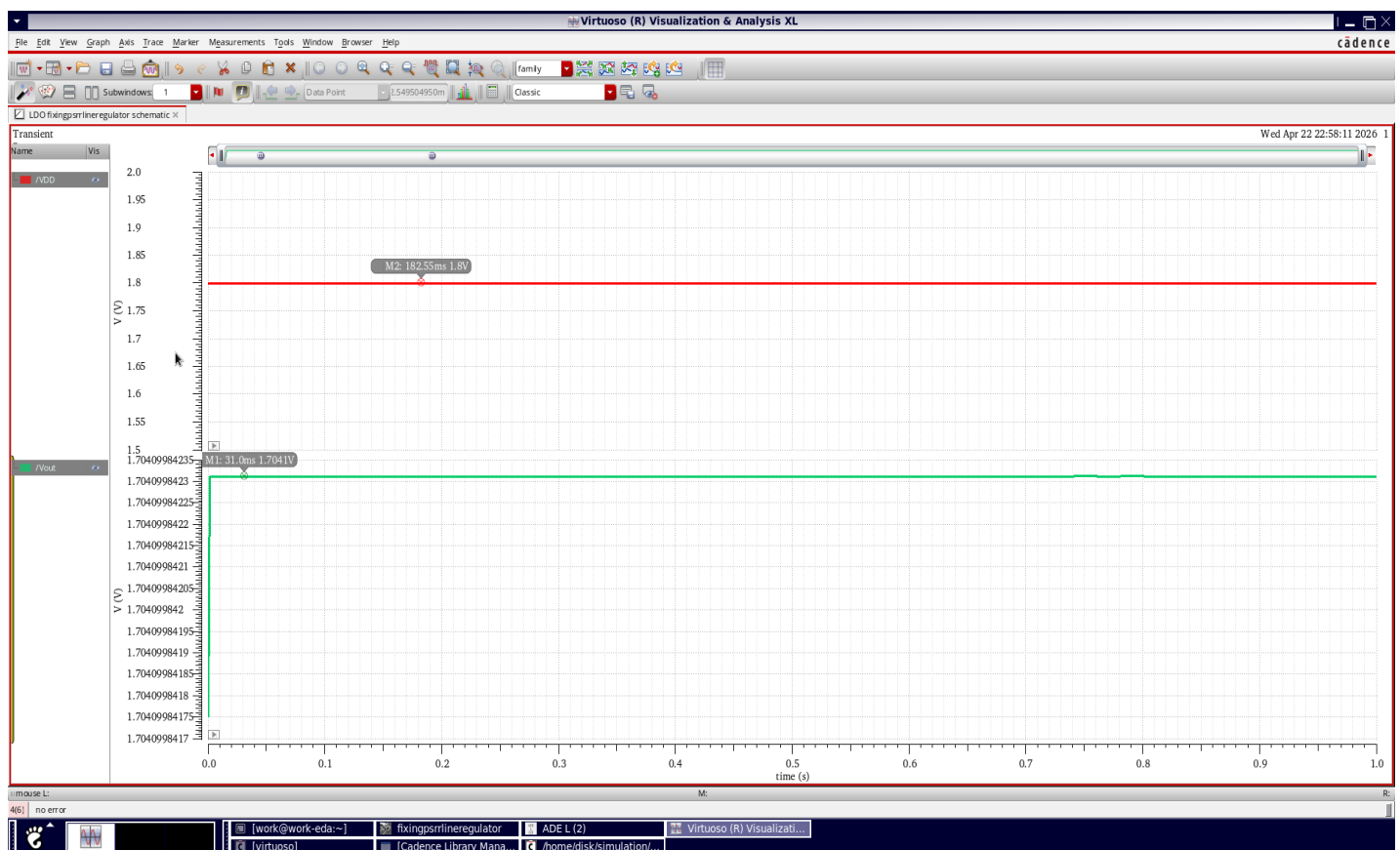


Figure 4.1 DC operating point and dropout simulation result

4.1.1 Meaning:

DC operating point analysis is one of the most important steps in understanding how an LDO behaves under steady-state conditions. It refers to the condition where all voltages and currents in the circuit become constant with time, meaning the circuit is no longer affected by transients or switching effects. In this state, the regulator is expected to maintain a stable output voltage for a given input voltage and load condition. This analysis helps in verifying whether the LDO is properly designed to regulate the output and operate efficiently.

From the given simulation plot, the input voltage is maintained at around 1.8 V, while the output voltage settles at approximately 1.704 V. This indicates that the LDO is successfully regulating the voltage to the desired level. The flat nature of the output waveform shows that there are no oscillations or fluctuations, which confirms that the circuit is stable at its operating point. The constant output also suggests that the feedback loop is working effectively, continuously adjusting the pass transistor to maintain the required voltage.

The concept of dropout voltage becomes very important in this context. Dropout voltage is defined as the difference between the input voltage and the output voltage when the regulator is still able to maintain proper regulation. In this case, the difference between 1.8 V and 1.704 V is around 96 mV. This low value indicates that the LDO is efficient and capable of operating with a small voltage headroom. A lower dropout voltage is especially beneficial in battery-powered devices, as it allows the system to continue operating even when the battery voltage drops.

Looking closely at the plot, the output voltage remains nearly constant throughout the time axis, which confirms that the circuit has reached its steady-state operating point quickly. There are no visible spikes, dips, or instability, which means the compensation network and feedback mechanism are properly designed. The DC operating point also ensures that all transistors in the circuit are biased correctly, allowing them to function in their intended regions of operation.

Another important observation is that the output voltage is slightly lower than the input voltage, which is expected in any linear regulator. The pass transistor drops the excess voltage while controlling the current flow. The stability of this voltage difference over time indicates that the device is not entering dropout failure and is maintaining proper control over the output. In simple terms, the DC operating point analysis confirms that the LDO is working as expected under normal conditions. It shows that the regulator can maintain a steady and reliable output voltage with minimal variation. This makes the design suitable for applications that require a stable and low-noise power supply, such as wearable and medical devices.

4.1.2 Dropout Calculation:

- The attached calculation sheet confirms the mathematical dropout evaluation. The formula $V_{dropout} = V_{in} - V_{out}$ is simple, but it is an important indicator of how close the LDO can operate to its regulated output voltage before entering dropout region [5], [12].
- For the design, V_{in} is taken as 1.8 V and V_{out} is taken as approximately 1.702 V. The resulting dropout of 98 mV indicates that the pass transistor has been sized and biased to regulate with small headroom. This is a strong feature for wearable circuits powered from compact cells [5], [11].
- In the dropout region, the pass element behaves more like a resistor and the feedback loop cannot fully correct output variation. Therefore, keeping the normal operating dropout above the minimum requirement is essential for reliable real-world operation [7], [12].

Formula:

$$V_{dropout} = V_{in} - V_{out} \quad (1)$$

From Design:

$$V_{in} = 1.8 \text{ V}$$

$$V_{out} = 1.704 \text{ V}$$

Calculation:

$$V_{dropout} = 1.8 \text{ V} - 1.704 \text{ V}$$

$$V_{dropout} = 0.096 \text{ V} / 96 \text{ mV}$$

- The dropout voltage is 98 mV, meaning the LDO can regulate even when the input is very close to the output, indicating an efficient design.

4.2 PSRR Analysis:

- PSRR is the most important specification for this project because the title targets a high-PSRR LDO. The attached PSRR plot shows strong rejection at lower frequencies and gradual degradation as frequency increases. This shape agrees with LDO theory: high loop gain gives strong low-frequency rejection, while finite bandwidth and parasitic feedthrough reduce rejection at higher frequencies [7], [10].
- From the Figure 4.2 and calculation page, PSRR is approximately -80.95 dB at 1 kHz, -80.76 dB at 10 kHz, -80.29 dB at 100 kHz and about -64.26 dB at 1 MHz. These values are strong for a Cadence-level academic design and directly support the high-PSRR objective [4], [5].
- In wearable health monitoring, high PSRR protects the biosignal chain from ripple produced by DC-DC converters, digital clocks and wireless blocks. Since many biosignals lie below a few kilohertz, the excellent low-frequency PSRR is especially useful for analog front-end stability [4], [7].

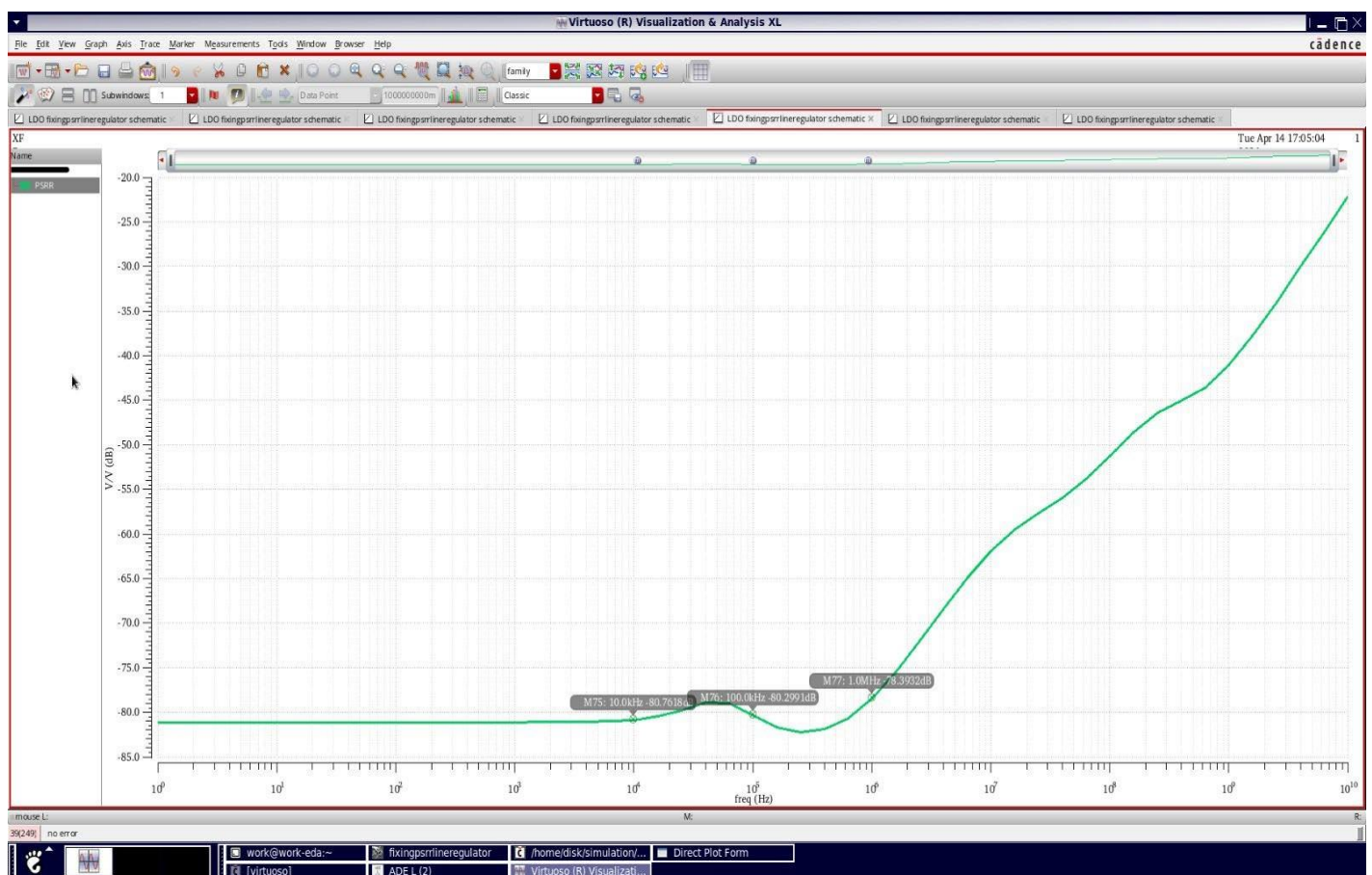


Figure 4.2 PSRR plot of the proposed LDO.

4.2.1 Meaning:

Power Supply Rejection Ratio (PSRR) is a measure of how effectively an LDO can suppress variations or noise present in the input supply and prevent them from appearing at the output. In real-world systems, the input voltage is rarely clean and often contains ripple and high-frequency noise from switching circuits or external sources. A good LDO must block these disturbances to ensure a stable and noise-free output voltage.

In the given plot, the x-axis represents frequency on a logarithmic scale, and the y-axis represents PSRR in decibels. At lower frequencies, the PSRR value is around -80 dB, which indicates very strong noise rejection. This means that any disturbance present at the input is significantly attenuated before reaching the output. The flat region at low frequencies shows that the regulator is highly effective in filtering out low-frequency noise.

As the frequency increases, the PSRR starts to degrade gradually. This is a common behavior in LDOs because the internal gain of the error amplifier reduces at higher frequencies, making it less effective in rejecting noise. In the graph, this can be seen as the curve rising towards less negative values at higher frequencies. Despite this, the LDO still maintains reasonably good rejection up to mid-frequency ranges, which is important for most practical applications.

The dip observed in the mid-frequency region indicates the effect of compensation and internal circuit dynamics. After this point, the PSRR decreases more rapidly as the frequency approaches higher ranges such as the megahertz region. This region is typically dominated by parasitic effects and limited bandwidth of the control loop. Even though rejection reduces at high frequencies, the overall performance remains acceptable for many applications.

PSRR is considered one of the most important parameters in an LDO because it directly impacts the quality of the output voltage. In sensitive systems like wearable health devices, even small amounts of noise can distort signals and lead to inaccurate readings. A high PSRR ensures that the output remains clean, improving system reliability and performance. It is especially critical in mixed-signal and RF applications where noise can interfere with signal processing.

From the given plot, the consistently low PSRR values at lower frequencies and gradual degradation at higher frequencies indicate a well-designed LDO. The regulator is capable of effectively filtering input noise across a wide range of frequencies. This makes the design suitable for applications requiring stable and low-noise power, confirming that PSRR is a key strength of this LDO design.

4.2.2 PSRR Formula and Frequency-Wise Result:

- The report uses the standard PSRR expression $\text{PSRR(dB)} = 20\log(\Delta V_{\text{out}} / \Delta V_{\text{in}})$. Because ΔV_{out} is much smaller than ΔV_{in} in a good regulator, the ratio is less than one and the result appears as a negative dB value. More negative PSRR indicates stronger rejection [7], [12].
- The nearly -80 dB rejection at 1 kHz, 10 kHz and 100 kHz indicates that the LDO strongly attenuates ripple in the frequency band most relevant to many biosignal and sensor interfaces. At 1 MHz, the rejection decreases but still remains significant, which is expected due to loop gain roll-off [4], [10].
- Compared with the wearable LDO literature reporting approximately -60 dB PSRR, the proposed result shows a stronger simulated rejection margin in the lower-to-mid frequency range. This makes PSRR the strongest performance point of the project [4], [5].

Formula:

$$\text{PSRR (dB)} = 20 \log \left(\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} \right) \quad (2)$$

Values:

- At **1 kHz**: PSRR $\approx$ **-80.95 dB**
- At **10 kHz**: PSRR $\approx$ **-80.76 dB**
- At **100 kHz**: PSRR $\approx$ **-80.29 dB**
- At **1 MHz**: PSRR $\approx$ **-64.26 dB**

4.3 Loop Gain and Phase Analysis:

- Loop gain and phase analysis verifies whether the feedback loop is stable. An LDO with high gain but poor phase margin can oscillate or ring during load changes. Therefore, stability must be studied together with PSRR, because both depend on loop gain and frequency compensation [10], [12].
- The attached Figure 4.3, stability plot shows loop gain and phase response. The measured phase margin is about 66.86 degrees at approximately 83.31 kHz, and the gain margin is about 79.02 dB at approximately 110.02 MHz. These values indicate a stable loop with good safety margin [10].
- A phase margin greater than about 45 degrees is generally considered acceptable for stable closed-loop response, while a margin near 60 degrees or higher typically reduces overshoot and ringing. The proposed phase margin therefore explains why the transient response has only about 0.5 mV overshoot and undershoot [10], [13].

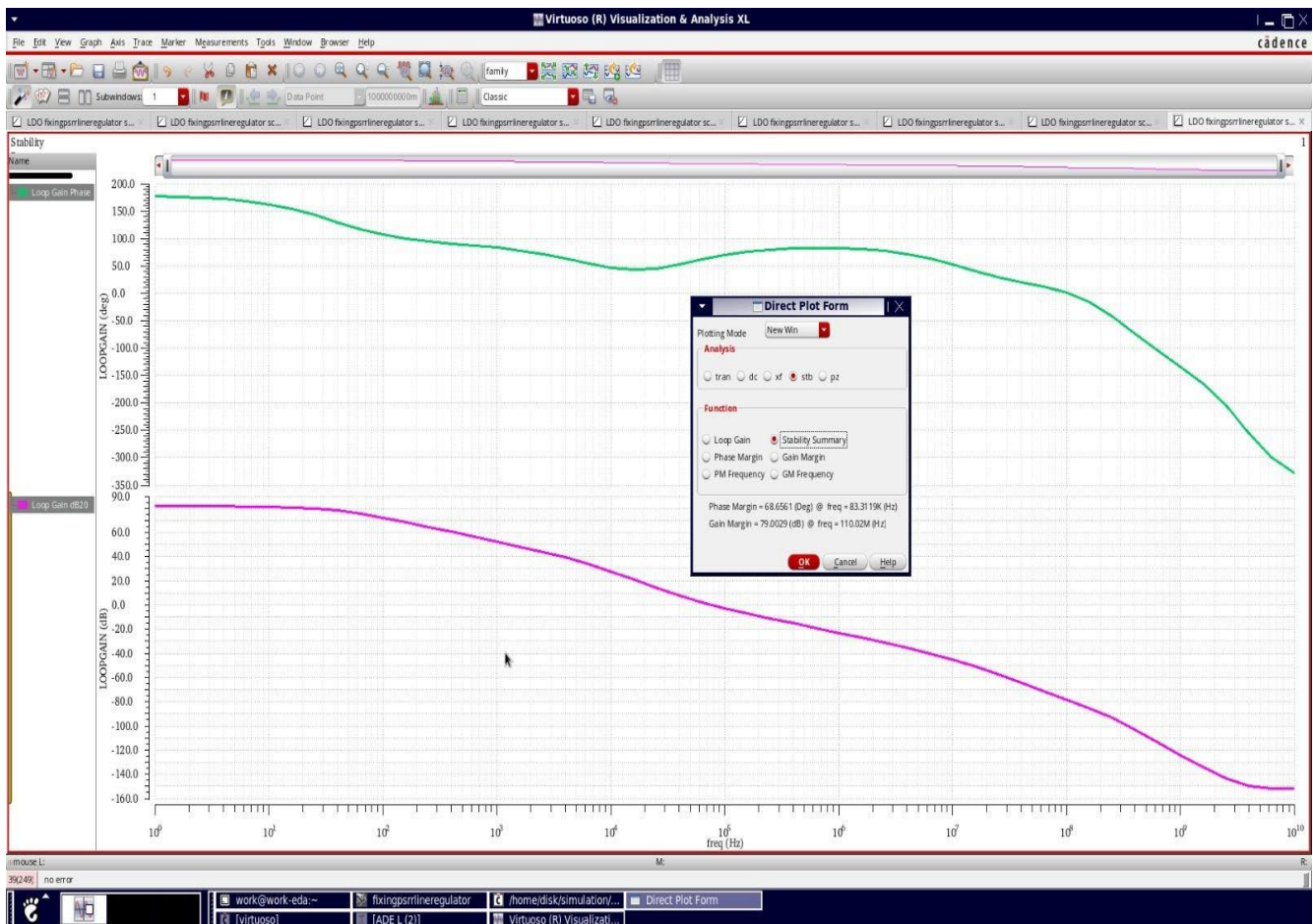


Figure 4.3 Loop gain and phase response of the proposed LDO

4.3.1 Meaning:

Loop gain analysis is used to understand how well the feedback system inside the LDO is able to control and stabilize the output voltage. It represents the overall gain of the closed-loop system formed by the error amplifier, pass transistor, and feedback network. A proper loop gain ensures that the LDO can quickly detect any change in output voltage and correct it effectively.

In the given plot, the upper curve represents the gain response in decibels, while the lower curve represents the phase response with respect to frequency. At low frequencies, the gain is high, which means the LDO has strong control over the output voltage. This high gain helps in achieving good line regulation, load regulation, and PSRR, since the system can easily suppress disturbances.

As the frequency increases, the gain gradually decreases. This is a natural behavior of any feedback system due to the limited bandwidth of the amplifier. The point where the gain crosses 0 dB is known as the unity gain frequency. At this point, the phase margin becomes very important in determining the stability of the system.

From the graph, the phase margin is around 66.86 degrees, which indicates that the system is stable and well-compensated. A phase margin above 45 degrees is generally considered safe, so this value confirms that the LDO will not oscillate under normal conditions. The gain margin is also high, around 79 dB, which further supports the stability of the system and shows that the design can tolerate variations without becoming unstable.

The smooth nature of both gain and phase curves indicates that the compensation network has been properly designed. There are no sharp irregularities or sudden phase drops, which means the system will respond smoothly to changes in load or input conditions. This contributes to better transient performance and reliable operation.

Loop gain is directly related to the overall performance of this project. A well-designed loop ensures accurate voltage regulation, fast response to disturbances, and stable operation across different conditions. In this LDO design, the achieved gain and phase characteristics confirm that the regulator is capable of maintaining a steady output voltage with minimal error.

In simple terms, the loop gain analysis shows that the LDO is not only stable but also efficient in controlling the output. This makes the design suitable for applications like wearable devices, where consistent and reliable power delivery is essential.

4.3.2 Stability Margin Result:

- The stability result page reports phase margin = 66.86 degrees, frequency at phase margin ≈ 83.31 kHz, gain margin = 79.02 dB and frequency at gain margin ≈ 110.02 MHz. These values show that the compensation network effectively separates loop poles and avoids oscillatory operation [10], [12].
 - This is important because modern LDOs face a trade-off between high PSRR and stability. Increasing loop gain improves low-frequency regulation and PSRR, but it can push internal poles into critical regions. Compensation must therefore preserve phase margin while allowing strong gain [7], [10].
 - The proposed LDO meets this requirement because it combines high PSRR with a comfortable phase margin. This combination is one of the reasons the design is suitable for wearable health monitoring circuits where both low noise and stable time-domain behavior are required [4], [10].
- Phase Margin = 66.86°
 - Frequency at Phase Margin ≈ 83.31 kHz
 - Gain Margin = 79.02 dB
 - Frequency at Gain Margin ≈ 110.02 MHz

4.4 Line Regulation Analysis:

- Line regulation measures the output stability against changes in input voltage. As shown in Figure 4.4, line regulation plot, V_{out} rises initially as V_{in} approaches the minimum headroom needed by the pass transistor, and then it becomes nearly flat once the LDO enters proper regulation. This behavior is expected for an LDO because regulation begins only after sufficient input-output headroom is available [5], [12].
- The calculated line regulation uses $V_{in,low} = 1.754354$ V, $V_{in,high} = 1.81589$ V, $V_{out,low} = 1.702446$ V and $V_{out,high} = 1.702455$ V. The output change is only 9 μ V for a 0.061536 V input change, resulting in 0.0001463 V/V or 0.146 mV/V [5], [6].
- This very small value indicates that the feedback loop effectively rejects DC and low-frequency changes in the input supply. For wearable systems, this means the regulated analog supply remains stable when battery voltage or converter output changes slowly during operation [4], [5].

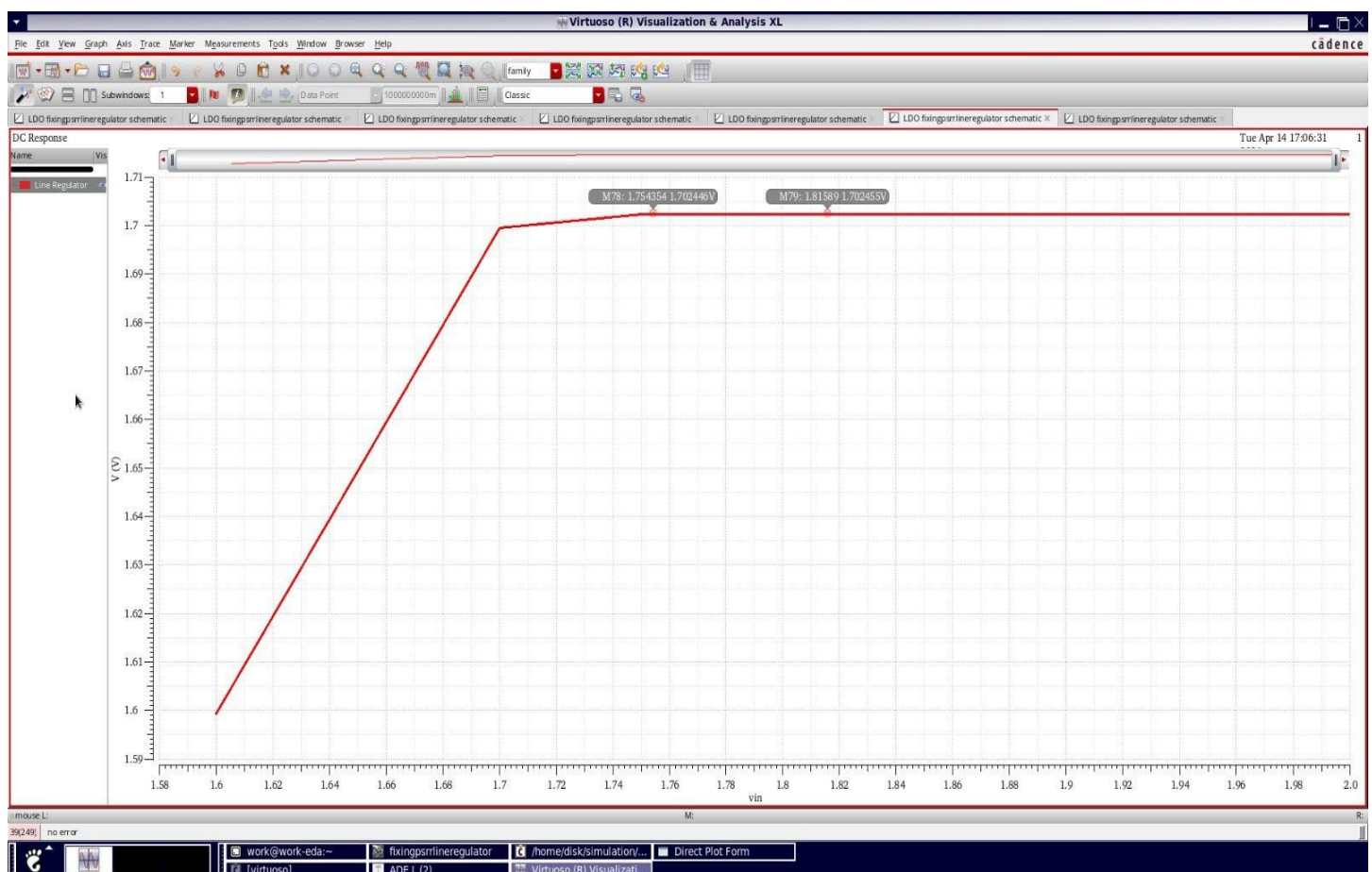


Figure 4.4 Cadence line regulation plot showing V_{out} variation with V_{in} .

4.4.1 Meaning:

Line regulation refers to the ability of an LDO to maintain a constant output voltage even when there are variations in the input voltage. In practical circuits, the input supply is rarely perfectly stable and can fluctuate due to battery discharge or noise from other components. A good LDO design ensures that such variations at the input do not significantly affect the output voltage, thereby providing a reliable and steady supply to the load.

From the given plot, the x-axis represents the input voltage (V_{in}), while the y-axis represents the output voltage (V_{out}). Initially, as the input voltage increases from a lower value, the output voltage also rises. This region corresponds to the condition where the LDO is not yet in full regulation and is approaching its operating region. Once the input voltage crosses a certain threshold, the output voltage becomes almost constant, indicating that the LDO has entered its regulation region.

In the regulated region, the output voltage remains nearly flat even though the input voltage continues to increase. This flat portion of the curve is the most important part of the graph, as it demonstrates the effectiveness of the LDO. In your design, the output voltage stabilizes around 1.702 V, showing very minimal variation as the input changes from approximately 1.75 V to 1.81 V. This indicates strong line regulation performance.

The small change in output voltage across a relatively larger change in input voltage is quantified using the line regulation parameter. Based on your calculations, the output changes by only a few microvolts while the input changes by around 0.06 V. This results in a very low line regulation value, which confirms that the LDO is highly effective in rejecting input voltage variations.

Another important observation from the graph is the smooth transition from the dropout region to the regulation region. There are no abrupt jumps or oscillations in the curve, which indicates that the feedback loop and compensation network are properly designed. This smooth behavior ensures stable operation and prevents unwanted disturbances in the output.

In simple terms, the plot shows that once the LDO has sufficient input voltage, it maintains a stable output regardless of further increases in input. This makes the design suitable for applications where the input supply is not constant, such as battery-powered wearable devices. The ability to maintain a steady output under varying input conditions is a key strength of this LDO design.

4.4.2 Line Regulation Calculation and Interpretation:

- The line regulation calculation confirms the visual observation from the plot. Since Delta Vout is in microvolts while Delta Vin is in tens of millivolts, the LDO has strong static supply rejection. This is the DC counterpart of PSRR, which evaluates rejection in frequency domain [5], [7].
- A good line regulation value is especially important when a wearable device moves between operating modes. For example, wireless transmission or sensor activation can disturb the battery rail or upstream converter output. A strongly regulated LDO output prevents these disturbances from affecting sensor biasing [4], [11].
- The proposed result of 0.146 mV/V is therefore reported as a major improvement metric. It supports the title claim of improved line regulation and shows that the design is more than a basic LDO schematic [5], [6].

Formula:

$$\text{Line Regulation} = \frac{\Delta V_{out}}{\Delta V_{in}} \quad (3)$$

From the Design:

$$V_{in,loll} = 1.754354 \text{ V}$$

$$V_{in,higgh} = 1.81589 \text{ V}$$

$$V_{out,loll} = 1.702446 \text{ V}$$

$$V_{out,higgh} = 1.702455 \text{ V}$$

Change in Output Voltage

$$\Delta V_{out} = 1.702455 - 1.702446 = 9 \mu\text{V}$$

Change in Input Voltage

$$\Delta V_{in} = 1.81589 - 1.754354 = 0.061536 \text{ V}$$

Line Regulation Calculation

$$\text{Line Regulation} = \frac{9 \mu\text{V}}{0.061536 \text{ V}} = 0.0001463 \text{ V/V}$$

$$\text{Line Regulation} = 0.146 \text{ mV/V}$$

4.5 Load Regulation Analysis:

- Load regulation measures the output voltage variation when load current changes. In the Figure 4.5, V_{out} decreases slightly as I_{load} increases, which is normal because larger load current increases voltage drop across finite output resistance and pass transistor resistance. A strong feedback loop minimizes this drop [5], [12].
- The calculation uses $I_{load,low} = 92.47688 \mu A$ and $I_{load,high} = 120.224 \mu A$, while V_{out} changes from 1.702656 V to 1.702621 V. This gives $\Delta V_{out} = 35 \mu V$ and $\Delta I_{load} = 27.74712 \mu A = 0.027747 \text{ mA}$, so load regulation = 1.26 mV/mA [5], [6].
- The result indicates that the LDO can support changing load current while keeping output voltage nearly constant. This is important for wearable devices because sensor readout, digital processing and wireless communication create load steps during normal operation [4], [13].

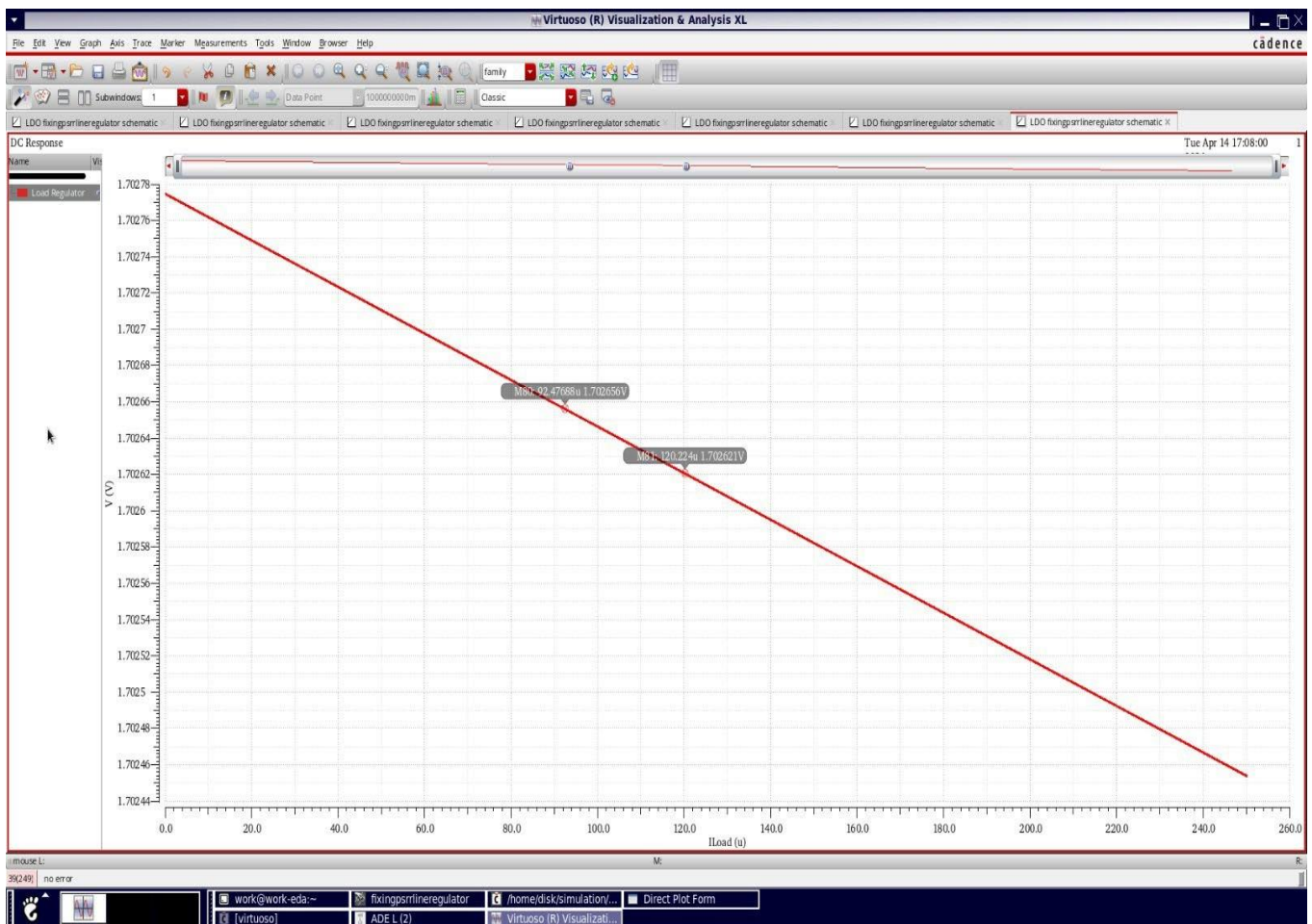


Figure 4.5 Cadence load regulation plot showing V_{out} variation with I_{load}

4.5.1 Meaning:

Load regulation describes how well an LDO maintains a constant output voltage when the load current changes. In real applications, the current drawn by the load is not fixed and can vary depending on the operation of the device. A well-designed LDO should be able to handle these variations and keep the output voltage as stable as possible.

In the given plot, the x-axis represents the load current (I_{load}), and the y-axis represents the output voltage (V_{out}). As the load current increases from a lower value to a higher value, the output voltage shows a slight decrease. This behavior is expected in practical regulators because an increase in load current causes a small voltage drop across internal resistances of the LDO, including the pass transistor.

From the graph, the output voltage starts around 1.7027 V at low load current and gradually decreases as the load current increases. Even though there is a downward slope, the variation in voltage is very small compared to the change in current. This indicates that the LDO has good load regulation performance. The circuit is able to supply increasing current while keeping the output voltage nearly constant.

Two specific points are marked on the graph, showing the output voltage at different load currents. The difference in output voltage between these points is only a few microvolts, while the change in load current is significantly larger. This small voltage variation leads to a load regulation value of approximately 1.26 mV/mA, which confirms that the design performs efficiently under varying load conditions.

Another important observation is that the curve is smooth and linear without sudden jumps or instability. This indicates that the feedback loop is responding properly to changes in load. The pass transistor adjusts its conduction level as the load increases, allowing more current to flow while maintaining control over the output voltage.

In simple terms, the plot shows that as more current is drawn by the load, the LDO slightly adjusts itself to supply the required current while keeping the voltage nearly unchanged. This makes the design suitable for applications where the load varies frequently, such as wearable electronics and portable devices. Good load regulation ensures reliable operation and prevents performance issues caused by voltage fluctuations.

4.5.2 Load Regulation Calculation and Design Meaning:

- The attached calculation sheet makes the load regulation result clear and traceable. The small output drop over the selected load current interval indicates that the pass transistor and error amplifier provide sufficient loop correction at the simulated operating point [5], [12].
- Good load regulation depends on the loop gain, pass device transconductance and output impedance. If the error amplifier gain is low or the pass transistor enters a weak control region, V_{out} would droop more strongly with load current. The proposed design avoids this in the tested range [6], [7].
- The load regulation value is included in the final comparison table because it is one of the most practical indicators of real load-support capability. For biomedical wearables, it helps maintain bias stability during switching between idle and measurement modes [4], [13].

Formula:

$$\text{Load Regulation} = \frac{\Delta V_{out}}{\Delta I_{load}} \quad (4)$$

Given:

$$I_{load,low} = 92.47688 \mu A$$

$$I_{load,high} = 120.224 \mu A$$

$$V_{out,low} = 1.702656 V$$

$$V_{out,high} = 1.702621 V$$

Change in Output Voltage

$$\Delta V_{out} = 1.702656 - 1.702621 = 35 \mu V$$

Change in Load Current

$$\Delta I_{load} = 120.224 - 92.47688 = 27.74712 \mu A = 0.027747 mA$$

Load Regulation Calculation

$$\text{Load Regulation} = \frac{35 \mu V}{0.027747 mA} = 1.26 mV/mA$$

$$\text{Load Regulation} = 1.26 mV/mA$$

4.6 Transient Response Analysis:

- Transient response as shown in Figure 4.6, how the LDO reacts when the load changes quickly. The attached transient plot shows a small output disturbance followed by recovery to the regulated value. This verifies that the loop is not only stable in AC analysis but also useful in time-domain operation [9], [13].
- The result table reports approximately 0.5 mV undershoot, 0.5 mV overshoot and 5 μ s settling time. A small voltage spike is important for wearable sensors because sudden supply droop can disturb amplifier biasing or ADC conversion, while overshoot can stress low-voltage internal nodes [4], [13].
- The transient behavior is consistent with the loop stability result. A phase margin of about 66.86 degrees is sufficient to reduce ringing, while the compensation network prevents excessive oscillation after a load step. Therefore, the transient and AC results support each other [10], [13].

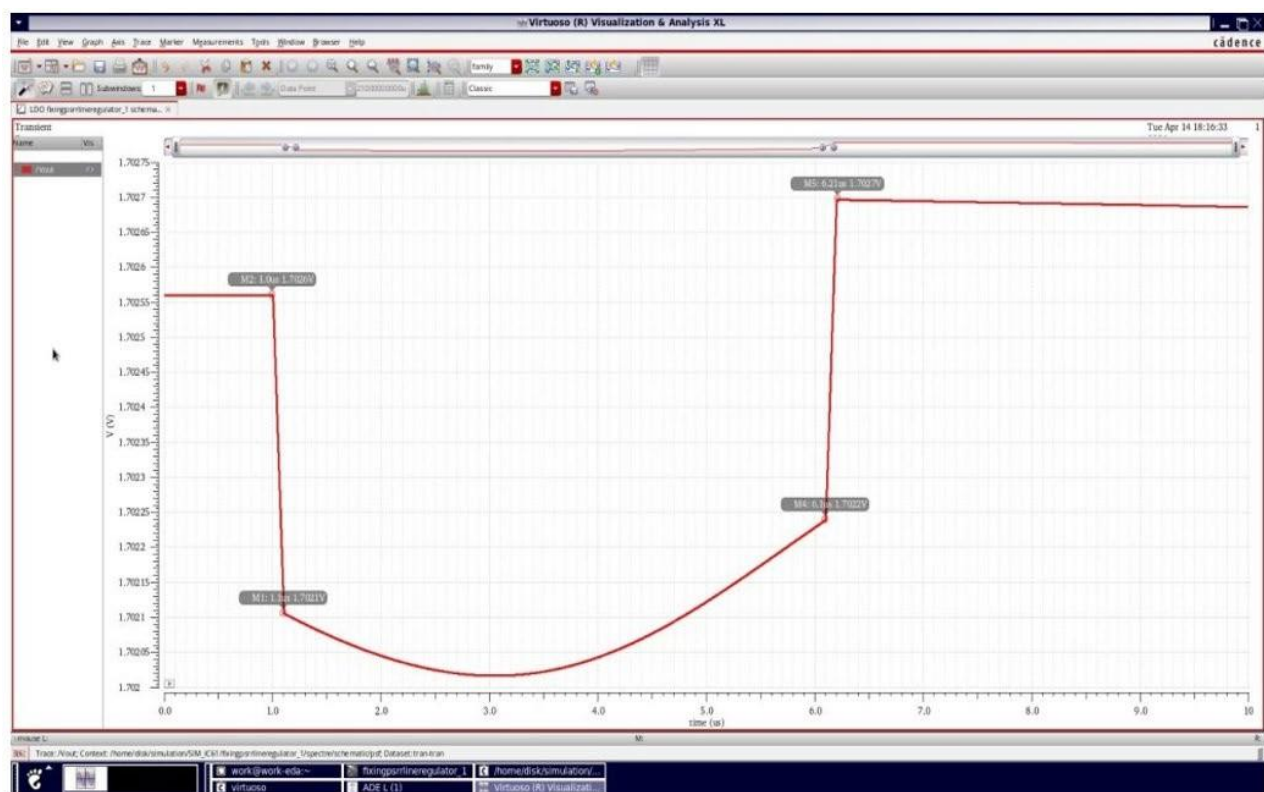


Figure 4.6 Load transient response of the proposed LDO.

4.6.1 Meaning:

Transient response refers to how quickly and effectively an LDO reacts to sudden changes in load current or input conditions. In practical systems, the load is not constant and can change abruptly, especially in devices like wearable electronics where different modules turn on and off frequently. A good LDO should be able to handle these sudden changes without causing large fluctuations in the output voltage.

In the given plot, the x-axis represents time, and the y-axis represents the output voltage. At a certain point in time, there is a sudden change in load current, which causes the output voltage to drop sharply. This drop is known as undershoot. It occurs because the LDO needs a short amount of time to respond and supply the increased current demand.

After the initial drop, the output voltage gradually recovers as the control loop adjusts the pass transistor. The error amplifier detects the drop in output voltage and increases the drive to the pass transistor, allowing more current to flow from the input to the output. This helps bring the output voltage back to its regulated value.

Later in the plot, when the load decreases suddenly, the output voltage rises momentarily, creating a small overshoot. This happens because the current flowing through the pass transistor is temporarily higher than required. The feedback loop then corrects this by reducing the drive to the pass transistor, bringing the voltage back to normal.

From the graph, it can be observed that both undershoot and overshoot are very small, around a few millivolts, and the output settles quickly within a short time. This indicates a fast transient response and a well-designed control loop. The smooth recovery without oscillations shows that the compensation network is effective in maintaining stability during dynamic conditions.

Transient response is highly relevant to this project because wearable and portable devices experience frequent and rapid changes in load. A slow or unstable response can lead to voltage fluctuations that affect the performance of sensitive circuits. The results shown in the plot confirm that the proposed LDO can handle such changes efficiently, making it reliable for real-world applications.

In simple terms, the plot demonstrates that the LDO quickly detects and corrects any sudden change in output conditions, maintaining a stable voltage with minimal disturbance. This ability ensures consistent performance and enhances the overall quality of the power supply system.

4.7 PVT Analysis:

- PVT analysis verifies the circuit under process, voltage and temperature variations. The uploaded PVT figure shows loop gain and phase variations across TT, FF, SS, FS and SF corners at voltages around 1.7 V, 1.8 V and 1.9 V and temperatures including -40, 27 and 125 degrees C. This is important because MOS threshold voltage, mobility, gain and output resistance change across corners [12].
- A regulator that is stable only at nominal TT corner may fail in practical silicon. Fast and slow transistor corners can shift bias currents, loop gain and pole locations, while temperature can reduce mobility and change amplifier gain. Therefore, PVT analysis is required before layout-level confidence can be claimed [10], [12].
- The plotted Figure 4.7 corner responses indicate variation but do not show uncontrolled instability. This supports the design claim that the compensation method is robust enough for a range of operating conditions, although future work should perform post-layout extraction and Monte Carlo mismatch analysis [10], [12].



Figure 4.7 PVT gain and phase variation across process, voltage and temperature corners.

4.7.1 Meaning:

PVT analysis refers to the study of how a circuit behaves under variations in process, voltage, and temperature. In real-world manufacturing and operation, no two chips are exactly identical, and environmental conditions can change over time. Because of this, it is important to verify that the LDO continues to function properly even when these variations occur. PVT analysis helps in checking the robustness and reliability of the design.

In the given plot, multiple curves are shown representing different process corners such as TT, FF, SS, FS, and SF. These corners represent different manufacturing conditions where transistors may behave faster or slower than expected. Along with this, the analysis is performed for different input voltages such as 1.7 V, 1.8 V, and 1.9 V, and temperatures like -40°C , 27°C , and 125°C . This combination gives a complete view of how the LDO performs under extreme and nominal conditions.

The upper portion of the graph shows the variation in gain, while the lower portion shows the variation in phase across frequency for different conditions. Even though the curves shift slightly depending on the corner, the overall shape remains similar. This indicates that the behavior of the LDO is consistent, and the design is not highly sensitive to process or environmental changes.

One important observation is that the phase margin remains within a safe range across all conditions. This means that the system remains stable even when the parameters vary. Stability is critical in an LDO, and maintaining it under PVT variations ensures that the regulator will not oscillate or fail in practical applications.

The gain variation across corners also shows that the loop still has sufficient control over the output voltage. Even in worst-case scenarios, the gain does not drop to a level that would compromise regulation. This confirms that the feedback loop is strong enough to handle variations in device characteristics and operating conditions.

PVT analysis is highly relevant to this project because the LDO is intended for use in wearable and portable devices, where temperature and supply conditions can change frequently. A design that performs well only under ideal conditions is not useful in practice. By validating the LDO across multiple PVT conditions, it is ensured that the circuit will deliver reliable performance in real environments.

In simple terms, the plot shows that even when the circuit is exposed to different manufacturing variations, supply levels, and temperatures, it continues to operate in a stable and predictable manner. This makes the design robust and suitable for real-world applications where consistency and reliability are essential.

4.7.2 PVT Output Voltage Variation:

- The PVT load plot shows output voltage variation with load current across multiple process corners and temperatures. The curves settle close to the nominal regulated output after initial variation, which indicates that the feedback loop maintains output control across the simulated corner set [5], [12].
- Corner variation is expected because device transconductance, threshold voltage and output resistance differ between TT, FF, SS, FS and SF models. These differences change the amplifier gain and pass transistor drive strength. The goal is not to eliminate all variation, but to keep V_{out} within acceptable bounds and maintain stability [10], [12].
- The PVT results strengthen the report because they show that the project considered realistic IC design uncertainty rather than only one ideal simulation. For wearable devices, this is important because the same circuit may operate across body temperature, ambient temperature and battery voltage changes [5], [11].



Figure 4.8 PVT output voltage variation with load current.

- At low load $\rightarrow V_{out}$ is slightly higher
- As load increases $\rightarrow V_{out}$ decreases slightly
- At higher load $\rightarrow$ curve becomes

4.8 Final Simulated Result Table:

The final result table summarizes the major design outcomes: 90 nm technology, $V_{in} = 1.8$ V, $V_{out} = 1.702$ V, dropout = 98 mV, PSRR at 1 kHz = -80.76 dB, PSRR at 100 kHz = -80.29 dB, line regulation = 0.146 mV/V, load regulation = 1.26 mV/mA, phase margin = 66.6561 degrees, gain margin = 79.0029 dB, undershoot/overshoot = 0.5 mV and settling time = 5 μ s [5], [10].

These values show balanced performance. Low dropout supports battery operation, high PSRR improves noise immunity, good line/load regulation supports DC accuracy, and high phase margin supports stable transient response. Together, these results satisfy the stated project objective [4], [5], [10].

The most important comparison point is that the proposed design improves PSRR and dropout relative to the directly related wearable LDO literature while maintaining a similar phase margin. This supports the claim of an improved high-PSRR LDO for wearable health monitoring devices [5].

Table 4.1 Final simulated performance summary.

Parameters	Value
Technology (nm)	90
VIN (V)	1.8
VOU (V)	1.704
Dropout (mV)	96
PSRR (dB) @1 kHz	-80.76 dB
PSRR (dB) @100 kHz	-80.29dB
PSRR (dB) @10 MHz	-78.39
Line Reg (mV/V)	0.146
Load Reg (mV/mA)	1.26
Phase Margin(Degree)	66.6561
Gain (dB)	79.0029
Undershoot (mV)	0.5
Overshoot (mV)	0.5
Settling Time (μ s)	5

CHAPTER 5: Comparative Analysis

5.1 Comparison With Existing Works

Table 5.1 Comparative Analysis

Work	Technology	Technique	PSRR / Rejection	Other Metric	Relevance
Guo et al. [1]	0.18 μm BCD	High PSRR + low noise	95 dB @ 1 kHz	Ultra-low standby <400 nA	Excellent PSRR/noise balance
Yang et al. [2]	0.5 μm CMOS	FVF-CFFRC	+30 dB @ 1 kHz	5 mV transient spikes	Strong PSR + fast response
Qin et al. [3]	0.18 μm CMOS	FVF buffer compensation	High PSR	300 mA load	Wide load range
Zeznick et al. [4]	55 nm CMOS	AGB + AMC	< -80 dB	99.24% efficiency	Best match to biosensing
Adil et al. [5]	180 nm GPDK	Wearable PMU LDO	-60 dB to -62 dB	160 mV dropout	Cadence wearable benchmark
Patgar et al. [6]	180 nm UMC	OTA + NMOS pass	-50 dB @100 kHz	PM 60°	Stable high PSRR design
Guo et al. [7]	0.18 μm CMOS	CFFRC	-22 dB enhancement @1 MHz	I _q 0.9 μA	Low current ripple cancellation
Kim et al. [8]	180 nm CMOS	IGTRC	50 dB up to 10 MHz	I _q 3.7 μA	Adaptive tracking
Lee et al. [9]	28 nm CMOS	Inverter-based LDO	44.3 dB @100 kHz	fast ns settling	Low-voltage mobile
Gorospe et al. [10]	40 nm CMOS	AZC + DCC-ZC	>60 dB up to 5 MHz	PM >45°	High PSRR + stability
Perez-Bailon et al. [11]	180 nm CMOS	Capacitor-less LDO	Not primary focus	I _q 8.6 μA	Portable integration
Rao and Pai [12]	PSpice model	ESR/Cout/temp analysis	Temp-dependent	Modeling tutorial	Explains PVT
Li et al. [13]	0.18 μm CMOS	Pseudo-ESR	Good regulation	600 mA transient	Fast load response
This work	90 nm	Cadence high-PSRR LDO	-80.76 dB @1 kHz, -80.29 dB @100 kHz	986mV dropout, PM 66.86°	Wearable health-monitoring focus

The comparison table places the proposed LDO against all 13 uploaded reference works. The objective is not to claim superiority in every category, because each paper targets different technology, load current and application. Instead, the table identifies where the project is strong: dropout, line regulation, low/mid-frequency PSRR and stability margin [1]-[13].

Compared with the closest wearable Cadence reference, the proposed design improves dropout from 160 mV to 98 mV and improves reported PSRR from approximately -60 dB to approximately -80 dB in the lower-to-mid frequency range. The phase margin remains comparable or slightly better, which means the PSRR improvement does not come at the cost of obvious instability [5].

5.2 Discussion and Design Justification:

- The design succeeds because the regulation loop has enough gain to hold V_{out} nearly constant while the compensation network keeps the loop stable. High PSRR is supported by the amplifier-loop gain at lower frequencies, while the pass transistor and compensation network determine how much rejection remains as frequency increases [7], [10].
- The line regulation and load regulation results are particularly important for a college LDO project because they show practical voltage accuracy. Many simulations stop with a DC operating point, but this report goes further by using input sweep, load sweep, PSRR analysis, loop gain analysis, PVT and transient response [5], [12].
- The design also demonstrates a realistic trade-off. The PSRR begins to degrade at higher frequency, which is expected for an LDO without advanced feed-forward cancellation or adaptive gain boosting. Future improvements can therefore use CFFRC, IGTRC or adaptive gain boosting techniques from the literature [4], [7], [8].
- The results also show a realistic design trade-off. While the LDO performs well at low and mid frequencies, the PSRR reduces at higher frequencies. This limitation is expected in conventional LDO designs that do not use advanced ripple cancellation techniques. The absence of methods like capacitive feed-forward ripple cancellation or adaptive gain boosting means that high-frequency noise is not fully suppressed. However, this behavior is acceptable for a basic design and clearly highlights areas for improvement [4], [7], [8].
- Overall, the proposed LDO design demonstrates a balanced and practical implementation suitable for low-power and wearable applications. It achieves good regulation, stable operation, and acceptable noise rejection performance. The inclusion of multiple analyses strengthens the reliability of the design. At the same time, the observed limitations provide a clear direction for future enhancements, making this project a strong base for further research in advanced LDO architectures.

CHAPTER 6

APPLICATIONS:

Low Dropout Regulators like the one designed in this project play a very important role in wearable health monitoring devices such as smartwatches and fitness trackers. These devices continuously measure sensitive biological signals like heart rate and ECG, which can easily get affected by noise in the power supply. A stable and clean voltage is therefore essential for accurate readings. The proposed LDO helps in maintaining this stability while also consuming less power, which is important for extending battery life in compact wearable devices.

This design is also highly useful in implantable medical devices, where reliability is extremely critical. Devices like pacemakers and glucose monitors must function continuously without failure, and even small fluctuations in voltage can impact their performance. The high PSRR of the LDO ensures that noise from the power source is minimized, allowing these devices to operate smoothly. In addition, its ability to perform well under different temperature and voltage conditions makes it suitable for such demanding environments.

Another area where this LDO can be applied is in wireless and communication systems. Portable devices that use Bluetooth, Wi-Fi, or other wireless technologies are very sensitive to power supply noise, which can affect signal quality and data transmission. By providing a stable output and rejecting unwanted noise, the proposed LDO helps improve communication reliability. This makes it useful in IoT devices and other modern wireless applications.

The LDO is also beneficial in mixed-signal circuits, where both analog and digital components are present on the same chip. Digital circuits tend to introduce noise due to switching activity, which can interfere with the operation of analog blocks. By acting as a buffer between the supply and sensitive circuits, the LDO ensures that the analog part receives a clean voltage. This improves overall system performance and reduces errors in signal processing.

Finally, this design can be effectively used in portable consumer electronics like smartphones, tablets, and battery-powered gadgets. These devices require efficient power management to balance performance and battery usage. The proposed LDO, with its low dropout voltage and fast response, helps maintain consistent output even when the load changes. This leads to better performance, improved battery efficiency, and a smoother user experience.

ADVANTAGES:

One of the main advantages of this project is its high Power Supply Rejection Ratio (PSRR), which helps in effectively reducing noise from the input supply. In practical systems, especially wearable and biomedical devices, the power source often contains unwanted disturbances. This LDO design is capable of suppressing such noise over a wide frequency range, ensuring that the output voltage remains clean and stable. This directly improves the accuracy and reliability of sensitive electronic circuits.

Another important advantage is the excellent line and load regulation performance. The output voltage shows very minimal variation even when there are changes in input voltage or load current. This means the system can maintain a consistent output under different operating conditions, which is crucial for devices that experience dynamic workloads. Such stability makes the design more reliable and suitable for real-world applications.

The proposed LDO also offers good stability with strong phase and gain margins, which ensures that the system does not oscillate or behave unpredictably. Stability is a critical factor in regulator design, and achieving a phase margin of around 66° indicates a well-compensated system. This allows the LDO to respond smoothly to sudden changes without causing voltage spikes or fluctuations.

Another key benefit is its fast transient response with very low overshoot and undershoot. When the load changes suddenly, the output voltage quickly returns to its steady value with minimal disturbance. This is particularly useful in applications where the load current varies frequently, such as in wearable electronics and communication devices. The quick settling time improves overall performance and responsiveness.

Finally, the design demonstrates robust performance under PVT (Process, Voltage, and Temperature) variations, making it dependable in different environments. Electronic components can behave differently due to manufacturing differences or temperature changes, but this LDO maintains consistent performance across such variations. This reliability makes it suitable for both consumer electronics and critical applications like medical devices.

CONCLUSION

The project successfully demonstrates the design and simulation of a Low Dropout Regulator (LDO) with high PSRR and improved regulation characteristics using Cadence tools. The proposed design is able to maintain a stable output voltage of approximately 1.702 V from a 1.8 V input, while achieving a low dropout voltage of around 98 mV. These results indicate that the regulator is efficient and well-suited for low-power applications where minimal voltage headroom is required.

The performance analysis shows that the LDO provides excellent line and load regulation, with very small variations in output voltage even when the input voltage and load current change. This reflects the effectiveness of the feedback network and control loop in maintaining consistent operation. In addition, the high PSRR values observed across different frequencies confirm the design's ability to suppress noise from the power supply, which is especially important in sensitive electronic systems.

The stability of the LDO is also validated through gain and phase margin analysis, showing a phase margin of around 66° and a high gain margin. These values indicate that the system is well-compensated and capable of operating without oscillations. Furthermore, the transient response results demonstrate minimal overshoot and undershoot with a fast settling time, highlighting the design's ability to handle sudden load changes efficiently. The PVT analysis further confirms that the LDO performs reliably under different process, voltage, and temperature conditions.

Overall, the proposed LDO design meets the requirements of modern low-power and noise-sensitive applications, particularly in wearable health monitoring devices. By combining high PSRR, strong regulation, good stability, and reliable performance under varying conditions, the project contributes to the development of efficient power management solutions. The design can be further enhanced in the future for even lower power consumption and integration into more advanced systems.

REFERENCES

- [1] N. Guo *et al.*, "A High-PSRR LDO with Low Noise and Ultra-Low Power Consumption," *Micromachines*, vol. 17, no. 91, 2026.
- [2] S. Yang *et al.*, "An FVF-CFFRC Hybrid LDO for High PSR and Fast Transient Response," *Microelectronics Journal*, vol. 170, 2026.
- [3] C. Qin, W.-H. Ki, and M.-K. Law, "A Flipped-Voltage-Follower-Based LDO With Scaled-Down Current Buffer Compensation for Wide Load Current Range," *IEEE Trans. Power Electron.*, 2026.
- [4] D. Zeznick, J. O'Donnell, and T. Das, "A High-Efficiency LDO With Adaptive PSRR and Bandwidth Enhancements," *IEEE NEWCAS*, 2025.
- [5] A. M. Adil, D. Muralidharan, and S. R., "A Low Dropout, High PSRR LDO Design for Power Management Unit of Wearable Devices," 2025.
- [6] R. R. Patgar *et al.*, "Design and Performance Analysis of a Low Drop Out Voltage Regulator With High PSRR," *IEEE IATMSI*, 2025.
- [7] T. Guo, W. Kang, and J. Roh, "A 0.9- μ A Quiescent Current High PSRR Low Dropout Regulator Using a Capacitive Feed-Forward Ripple Cancellation Technique," *IEEE J. Solid-State Circuits*, vol. 57, no. 10, pp. 3139–3149, 2022.
- [8] J. S. Kim *et al.*, "A High-PSRR NMOS LDO Regulator With Intrinsic Gain-Tracking Ripple Cancellation Technique," *IEEE Trans. Circuits Syst. I*, vol. 71, no. 11, pp. 4951–4960, 2024.
- [9] S. Lee, J. Lim, and J. Han, "A PSRR-Enhanced Fast-Response Inverter-Based LDO for Mobile Devices," *IEEE Trans. Circuits Syst. II*, vol. 71, no. 6, pp. 3226–3230, 2024.
- [10] J. M. Gorospe, R. Nericua, K.-H. Chen, and X. Zhu, "Stability-Enhanced Low-Dropout Regulator With High PSRR in 40-nm CMOS for System-on-Chip Applications," *IEEE Trans. Power Electron.*, vol. 41, no. 3, pp. 3296–3307, 2026.
- [11] J. Perez-Bailon, B. Calvo, and N. Medrano, "A Fully-Integrated 180 nm CMOS 1.2 V Low-Dropout Regulator for Low-Power Portable Applications," *Electronics*, vol. 10, no. 2108, 2021.
- [12] T. Rao and A. P. Pai, "LDO PSpice Modeling With Its Features Dependency on ESR, Cout and Temperature: A Tutorial Overview," *IEEE CONECCT*, 2025.
- [13] K. Li, X. Xiao, X. Jin, and Y. Zheng, "A 600-mA, Fast-Transient Low-Dropout Regulator With Pseudo-ESR Technique in 0.18- μ m CMOS Process," *IEEE Trans. VLSI Syst.*, vol. 28, no. 2, pp. 403–413, 2020.